

Definite Integration

Question1

$$\int_0^1 x \sin^{-1} x dx =$$

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Options:

A.

$$\frac{\pi}{8}$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{12}$$

D.

$$\frac{\pi}{3}$$

Answer: A

Solution:

$$I = \int_0^1 x \sin^{-1} x \cdot dx$$

$$\therefore \int uv \cdot dx = u \int v dx - \int \left(u' \int u dx \right) dx$$

Here, $u = \sin^{-1} x, v = x$

$$\begin{aligned}
\therefore I &= \left[\sin^{-1} x \int x \cdot dx - \int \left(\frac{1}{\sqrt{1-x^2}} \int x \cdot dx \right) dx \right]_0^1 \\
&= \left[\sin^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2\sqrt{1-x^2}} dx \right]_0^1 \\
&= \left(\sin^{-1}(1) \cdot \frac{1}{2} - \sin^{-1}(0) \cdot 0 \right) - \int_0^1 \frac{x^2}{2\sqrt{1-x^2}} dx \\
&= \frac{\pi}{4} - \frac{1}{2} \int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx
\end{aligned}$$

Let $x = \sin \theta \Rightarrow dx = \cos \theta \cdot d\theta$ at $x = 1 \Rightarrow \theta = \pi/2$ and $x = 0 \Rightarrow \theta = 0$

$$\begin{aligned}
&= \frac{\pi}{4} - \frac{1}{2} \int_0^{\pi/2} \frac{\sin^2 \theta \cdot \cos \theta}{\cos \theta} d\theta \\
&= \frac{\pi}{4} - \frac{1}{2} \int_0^{\pi/2} \sin^2 \theta \cdot d\theta \\
&= \frac{\pi}{4} - \frac{1}{2} \int_0^{\pi/2} \left(\frac{1}{2} - \frac{\cos 2\theta}{2} \right) d\theta \\
&= \frac{\pi}{4} - \frac{1}{2} \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/2} \\
&= \frac{\pi}{4} - \frac{1}{2} \left[\frac{\pi}{4} - 0 - 0 \right] = \frac{\pi}{4} - \frac{\pi}{8} = \frac{\pi}{8}
\end{aligned}$$

Question2

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin(x - [x]) dx =$$

Here $[x]$ is the greatest integer function

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Options:

A.

0

B.

$3(1 - \cos 1) + \sin 2 - \sin 1$

C.

$$3(1 - \cos 1) + \cos 2 - \sin 1$$

D.

$$\cos 2 - \sin 2$$

Answer: B

Solution:

$$\therefore I = \int_{-\pi/2}^{\pi/2} \sin(x - [x]) dx$$

$$\text{for } x \in [-\pi/2, -1] \Rightarrow [x] = -2$$

$$\text{and } x - [x] = x + 2$$

$$\text{for } x \in [-1, 0) \Rightarrow [x] = -1$$

$$\text{and } x - [x] = x + 1$$

$$\text{for } x \in [0, 1] \Rightarrow [x] = 0 \text{ and } x - [x] = x$$

$$\text{for } x \in [1, \frac{\pi}{2}] \Rightarrow x - [x] = x - 1$$

$$\therefore I = \int_{-\pi/2}^{-1} \sin(x + 2) dx + \int_{-1}^0 \sin(x + 1) dx + \int_0^1 \sin x \cdot dx + \int_1^{\pi/2} \sin(x - 1) dx$$

$$\begin{aligned} &= [-\cos(x + 2)]_{-\pi/2}^{-1} + [-\cos(x + 1)]_{-1}^0 + [-\cos x]_0^1 + [-\cos(x - 1)]_1^{\pi/2} \\ &= -\cos(1) + \cos(2 - \pi/2) - \cos(1) + \cos(0) - \cos(1) + \cos(0) - \cos(\pi/2 - 1) + \cos(0) \\ &= -3\cos(1) + 3 + \cos(-(\pi/2 - 2)) - \cos(\pi/2 - 1) \\ &= 3 - 3\cos(1) + \sin(2) - \sin(1) \end{aligned}$$

Question3

$$\int_0^2 x^2(2 - x)^5 dx =$$

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Options:

A.

$$\frac{128}{21}$$

B.

$$\frac{64}{7}$$

C.

$$\frac{32}{21}$$

D.

$$\frac{16}{7}$$

Answer: C

Solution:

$$\begin{aligned} & \int_0^2 x^2(2-x)^5 dx \\ &= \int_0^2 x^2 (32 - 80x + 80x^2 - 40x^3 + 10x^4 - x^5) dx \\ &= \left[\frac{32x^3}{3} - \frac{80x^4}{4} + \frac{80x^5}{5} - \frac{40x^6}{6} + \frac{10x^7}{7} - \frac{x^8}{8} \right]_0^2 \\ &= \left(\frac{32 \times 8}{3} - 20 \times 16 + 16 \times 32 - \frac{20}{3} \times 64 + \frac{10}{7} \times 128 - \frac{256}{8} \right) - 0 \\ &= \frac{256}{3} + 16 \times 12 - \frac{1280}{3} + \frac{1280}{7} - 32 \\ &= \frac{-1024}{3} + \frac{1280}{7} + 160 \\ &= \frac{-7168 + 3840 + 3360}{21} = \frac{32}{21} \end{aligned}$$

Question4

If $f(x) = \max \{x^3 - 4, x^4 - 4\}$ and $g(x) = \min \{x^2, x^3\}$, then $\int_{-1}^1 (f(x) - g(x)) dx =$

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Options:

A.

$$-\frac{151}{20}$$

B.

$$\frac{9}{20}$$

C.

$$\frac{131}{22}$$

D.

$$-\frac{67}{9}$$

Answer: A

Solution:

$$\because f(x) = \max \{x^3 - 4, x^4 - 4\},$$

$$g(x) = \min \{x^2, x^3\}$$

$$\text{at } x = -1 \Rightarrow x^2 = 1, x^3 = -1$$

$$\Rightarrow \min\{1, -1\} = -1 = (-1)^3$$

$$\text{Similarly, at } x = 0.5, \Rightarrow x^2 = 0.25,$$

$$x^3 = 0.125$$

$$\Rightarrow \min\{0.25, 0.125\} = 0.125 = (0.5)^3$$

$$\text{So, for } x \in [-1, 1] \Rightarrow g(x) = x^3$$

Similarly, analysis of $f(x)$

$$\text{When, } x^3 - 4 = x^4 - 4$$

$$\Rightarrow x^3(x - 1) = 0$$

$$\text{So, } x = 0, 1$$

$$\text{for } x < 0 \Rightarrow x^4 - 4 > x^3 - 4$$

$$\text{for } 0 < x < 1$$

$$\text{thus, } f(x) = \begin{cases} x^4 - 4, & -1 \leq x \leq 0 \\ x^3 - 4, & 0 < x \leq 1 \end{cases}$$

Therefore, $f(x) - g(x)$

$$= \begin{cases} (x^4 - 4) - x^3 = x^4 - x^3 - 4, & -1 \leq x \leq 0 \\ (x^3 - 4) - x^3 = -4, & 0 < x \leq 1 \end{cases}$$

$$\therefore \int_{-1}^1 (f(x) - g(x)) dx = \int_{-1}^0 (x^4 - x^3 - 4) dx + \int_0^1 -4 \cdot dx$$

$$\begin{aligned}
&= \left[\frac{x^5}{5} - \frac{x^4}{4} - 4x \right]_{-1}^0 - 4[x]_0^1 \\
&= \left[0 - \left(-\frac{1}{5} - \frac{1}{4} + 4 \right) - 4(1) - 0 \right] \\
&= -8 + \frac{1}{5} + \frac{1}{4} = \frac{-160+5+4}{20} = \frac{-151}{20}
\end{aligned}$$

Question5

$$\int_0^\pi (\sin^5 x \cos^3 x + \sin^4 x \cos^4 x + \sin^3 x \cos^4 x) dx =$$

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Options:

A.

$$\frac{873}{2240}$$

B.

$$\frac{3\pi}{128} + \frac{12}{35}$$

C.

$$\frac{1641}{4480}$$

D.

$$\frac{3\pi}{128} + \frac{4}{35}$$

Answer: D

Solution:

$$\begin{aligned}
\text{Let } I &= \int_0^\pi (\sin^5 x \cos^3 x + \sin^4 x \cos^4 x + \sin^3 x \cos^4 x) dx \quad \dots (i) \\
&\Rightarrow I = \int_0^\pi [\sin^5(\pi - x) \cos^3(\pi - x) \\
&\quad + \sin^4(\pi - x) \cos^4(\pi - x) \\
&\quad + \sin^3(\pi - x) \cdot \cos^4(\pi - x)] dx \\
&\Rightarrow I = \int_0^\pi -\sin^5 x \cos^3 x + \sin^4 x \cos^4 x \\
&\quad + \sin^3 x \cos^4 x dx \quad \dots (ii)
\end{aligned}$$

On adding Eqs. (i) and (ii), we get

$$\begin{aligned}
2I &= 2 \int_0^\pi (\sin^4 x \cos^4 x + \sin^3 x \cos^4 x) dx \\
I &= \int_0^\pi \frac{\sin^4 2x}{16} dx + \int_0^\pi \sin^3 x \cdot \cos^4 x dx
\end{aligned}$$

$$\begin{aligned}
\cos x &= t \\
&\Rightarrow -\sin x dx = dt \\
&= \frac{1}{16} \int_0^\pi \frac{(1 - \cos 4x)^2}{4} dx + \int_1^{-1} t^4 \cdot (1 - t^2)(-dt) \\
&= \frac{1}{64} \int_0^\pi (1 - 2 \cos 4x + \cos^2 4x) dx + \int_1^{-1} t^4 (t^2 - 1) dt \\
&= \frac{1}{64} \int_0^\pi \left(1 - 2 \cos 4x + \frac{1 + \cos 8x}{2} \right) dx + \left(\frac{t^7}{5} - \frac{t^5}{5} \right)_1^{-1} \\
&= \frac{1}{64} \int_0^\pi \left(\frac{3}{2} - 2 \cos 4x + \frac{1}{2} \cos 8x \right) dx + \left(\frac{2}{5} - \frac{2}{7} \right) \\
&= \frac{3}{128} [x]_0^\pi - \frac{2}{64} \left[\frac{\sin 4x}{4} \right]_0^\pi + \frac{1}{128} \left[\frac{\sin 8x}{8} \right]_0^\pi + \frac{4}{35} \\
&\Rightarrow \frac{3\pi}{128} + \frac{4}{35}
\end{aligned}$$

Question6

$$\int_0^1 \frac{x^4+1}{x^6+1} dx =$$

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Options:

A.

$$\frac{\pi}{3}$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{6}$$

D.

$$\frac{\pi}{2}$$

Answer: A

Solution:

$$I = \int_0^1 \frac{x^4 + 1}{x^6 + 1} dx$$

$$\text{put } x^6 + 1 = (x^2)^3 + 1$$

$$= (x^2 + 1)(x^4 - x^2 + 1)$$

$$= \int_0^1 \frac{(x^4 + 1)}{(x^2 + 1)(x^4 - x^2 + 1)} dx$$

$$\Rightarrow \frac{x^4 + 1}{(x^2 + 1)(x^4 - x^2 + 1)}$$

$$= \frac{A}{x^2 + 1} + \frac{Bx^2 + C}{x^4 - x^2 + 1}$$

$$x^4 + 1 = A(x^4 - x^2 + 1) + Bx^2(x^2 + 1) + Cx^2 + C$$

$1 = A + B$	$-A + B + C = 0$	$A + C = 1$
$B = 1 - A$	$A - B - C = 0$	$C = 1 - A$

$$\therefore A - (1 - A) - (1 - A) = 0$$

$$\Rightarrow A - 1 + A - 1 + A = 0$$

$$\Rightarrow 3A = 2$$

$$\Rightarrow A = 2/3$$

$$B = C = 1 - 2/3 = 1/3$$

$$I = \frac{2}{3} \int_0^1 \frac{1}{x^2 + 1} dx - \frac{1}{3} \int \frac{x^2 + 1}{x^4 - x^2 + 1} dx$$

$$= \frac{2}{3} [\tan^{-1} x]_0^1 - \frac{1}{3} \int_0^1 \frac{1 + 1/x^2}{x^2 + 1/x^2 - 1} dx$$

$$= \frac{2}{3} \left(\frac{\pi}{4} \right) - \frac{1}{3} \int_0^1 \frac{(1 + 1/x^2)}{(x - \frac{1}{x})^2 + 1} dx$$

$$= \frac{\pi}{6} - \frac{1}{3} [\tan^{-1}(x - 1/x)]_0^1$$

$$= \frac{\pi}{6} - 1/3 \left(\frac{-\pi}{2} \right)$$

$$\Rightarrow \frac{\pi}{6} + \pi/6 = \frac{\pi}{3}$$

Question7

$$\int_{-2\pi}^{2\pi} \sin^4(2x) \cos^6(2x) dx =$$

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Options:

A.

$$\frac{3\pi}{64}$$

B.

$$\frac{9\pi}{64}$$

C.

$$\frac{9\pi}{35}$$

D.

$$\frac{9\pi}{280}$$

Answer: A

Solution:

$$I = \int_{-2\pi}^{2\pi} \sin^4(2x) \cos^6(2x) dx$$

Let $u = 2x$, then $du = 2dx$,

$$\text{so } dx = \frac{1}{2} du$$

When $x = -2\pi$, $u = 2(-2\pi) = -4\pi$

when $x = 2\pi$, $u = 2(2\pi) = 4\pi$

$$\begin{aligned} I &= \int_{-2\pi}^{2\pi} \sin^4(2x) \cos^6(2x) dx \\ &= \frac{1}{2} \int_{-4\pi}^{4\pi} \sin^4(u) \cos^6(u) du \end{aligned}$$

Since, $\sin(u)$ and $\cos(u)$ are periodic with a period of 2π ,

So, $\sin^4(u)$ and $\cos^6(u)$ are also periodic with a period of 2π

... (ii)

$$\begin{aligned}
 &= \frac{1}{2} \cdot 2 \int_{-2\pi}^{2\pi} \sin^4(u) \cos^6(u) du \\
 &= \int_{-2\pi}^{2\pi} \sin^4(u) \cos^6(u) du \\
 &= \frac{1}{2} \cdot 4 \int_0^{2\pi} \sin^4(u) \cos^6(u) du \\
 &= 2 \int_0^{2\pi} \sin^4(u) \cos^6(u) du
 \end{aligned}$$

Since, the function, $f(u) = \sin^4(u) \cos^6(u)$ is an even function.

$$\begin{aligned}
 \therefore 2 \int_0^{2\pi} \sin^4(u) \cos^6(u) du \\
 &= 2 \cdot 2 \int_0^{\pi} \sin^4(u) \cos^6(u) du \\
 &= 4 \int_0^{\pi} \sin^4(u) \cos^6(u) du \\
 &= 4 \cdot 2 \int_0^{\pi/2} \sin^4(u) \cos^6(u) du \\
 &= 8 \int_0^{\pi/2} \sin^4(u) \cos^6(u) du
 \end{aligned}$$

We know that, $\int_0^{\pi/2} \sin^m(u) \cos^n(u) du$

$$= \frac{(m-1)(m-3)\dots(1 \text{ or } 2)(n-1)(n-3)\dots(1 \text{ or } 2)}{(m+n)(m+n-2)\dots(1 \text{ or } 2)} \times \alpha$$

where, $\alpha = \frac{\pi}{2}$ if both m and n are even, and $\alpha = 1$ otherwises.

Here, $m = 4, n = 6$ both are even.

So, $\int_0^{\pi/2} \sin^4(u) \cos^6(u) du$

$$\begin{aligned}
 &= \frac{(4-1)(4-3)(6-1)(6-3)(6-5)}{(4+6)(4+6-2)(4+6-4)} \cdot \frac{\pi}{2} \\
 &\quad (4+6-6)(4+6-8) \\
 &= \frac{3 \cdot 1 \cdot 5 \cdot 3 \cdot 1}{10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{45}{3840} \cdot \frac{\pi}{2} = \frac{3}{256} \cdot \frac{\pi}{2} \\
 &\quad = \frac{3\pi}{512} \\
 \therefore 8 \int_0^{\pi/2} \sin^4(u) \cos^6(u) du \\
 &= 8 \cdot \frac{3\pi}{512} = \frac{24\pi}{512} = \frac{3\pi}{64} \\
 \therefore 8 \int_0^{\frac{\pi}{2}} \sin^4(u) \cos^6(u) du &= \frac{3\pi}{64}
 \end{aligned}$$

Question 8

If $f(t) = \int_0^t \tan^{(2n-1)} x dx, n \in N$, then $f(t + \pi) =$

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Options:

A.

$$f(t)f(\pi)$$

B.

$$f(t) - f(\pi)$$

C.

$$f(t) + f(\pi)$$

D.

$$\frac{f(t)}{f(\pi)}$$

Answer: C

Solution:

$$\text{Given, } f(t) = \int_0^t \tan^{(2n-1)} x dx, n \in$$

$$\text{So, } f(t + \pi) = \int_0^{t+\pi} \tan^{(2n-1)} x dx$$

$$= \int_0^t \tan^{(2n-1)} x dx + \int_t^{t+\pi} \tan^{(2n-1)} x dx$$

$$\text{Let } u = x - \pi$$

$$\Rightarrow du = dx$$

Limit, when $x = t, u = t - \pi$ and

$$x = t + \pi, u = t$$

$$\text{So, } f(t + \pi) = f(t) + \int_t^{t+\pi} \tan^{(2n-1)}(u + \pi) du$$

$$\text{Since, } \tan(u + \pi) = \tan u$$

$$= f(t) + \int_{t-\pi}^t \tan^{(2n-1)}(u) du$$

$$= f(t) + \int_{t-\pi}^0 \tan^{(2n-1)}(u) du + \int_0^t \tan^{(2n-1)}(u) du$$

$$\text{Let } v = -u, \text{ the } dv = -du$$

When $u = t - \pi, v = \pi - t$, when $u = 0, v = 0$, so we get

$$\begin{aligned}
&= f(t) - \int_{\pi-t}^0 \tan^{(2n-1)}(-v)dv + \int_0^t \tan^{(2n-1)}(u)du \\
&= f(t) + \int_{\pi-t}^0 \tan^{(2n-1)}(v)dv + \int_0^t \tan^{(2n-1)}(u)du
\end{aligned}$$

$$\begin{aligned}
&= f(t) + \int_{\pi-t}^0 \tan^{(2n-1)}(v)dv + f(t) \\
&= 2f(t) - \int_0^{\pi-t} \tan^{(2n-1)}(v)dv \\
&= 2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv - \int_{\pi-t}^{\pi} \tan^{(2n-1)}(v)dv
\end{aligned}$$

Let $w = v - \pi$, then $dw = dv$

When $v = \pi - t$, $w = -t$, when $v = \pi$

$w = 0$

$$\begin{aligned}
&= 2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv - \int_{-t}^0 \tan^{(2n-1)}(w + \pi)dw \\
2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv - \int_{-t}^0 \tan^{(2n-1)}(w)dw
\end{aligned}$$

Let $z = -w$, then $dz = -dw$ when $w = -t$, $z = t$ and $w = 0$, $z = 0$

$$\begin{aligned}
&= 2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv + \int_t^0 \tan^{(2n-1)}(-z)dz \\
&= 2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv - \int_0^t \tan^{(2n-1)}(z)dz \\
&= 2f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv - f(t) \\
&= f(t) - \int_0^{\pi} \tan^{(2n-1)}(v)dv \\
&= f(t) + 0 \left[\because \int_0^{\pi} \tan^{(2n-1)}(v)dv = 0 \right] \\
&\quad = f(t) + f(\pi) \quad [\because f(\pi) = 0] \\
\therefore f(t + \pi) &= f(t) + f(\pi)
\end{aligned}$$

Question9

$$\int_0^2 x^8 \left(\frac{4}{x^2} - 1 \right)^{\frac{5}{2}} dx =$$

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Options:

A.

$$\frac{2^{15}}{63}$$

B.

$$\frac{2^{16}}{315}$$

C.

$$\frac{2^{16}}{189}$$

D.

$$\frac{2^{10}}{63}$$

Answer: D

Solution:

$$\begin{aligned} I &= \int_0^2 x^8 \left(\frac{4}{x^2} - 1 \right)^{5/2} dx \\ &= \int_0^2 x^8 \left(\frac{4 - x^2}{x^2} \right)^{5/2} dx \\ &= \int_0^2 x^8 \cdot \frac{(4 - x^2)^{5/2}}{x^5} dx \\ &= \int_0^2 x^3 (4 - x^2)^{5/2} dx \end{aligned}$$

Let $u = 4 - x^2$

Then, $du = -2x dx$, so $x dx = \frac{-1}{2} du$ and $x^3 = x \cdot x^2 = x(4 - u)$ when $x = 0, u = 4$ and when $x = 2, u = 0$

$$\begin{aligned}
& \therefore \int_0^2 x^8 \left(\frac{4}{x^2} - 1 \right)^{5/2} dx \\
&= \int_0^2 x^3 (4 - x^2)^{5/2} dx \\
&= \int_4^0 (4 - u) \cdot u^{5/2} \cdot \left(\frac{-1}{2} \right) du \\
&= \frac{1}{2} \left[4 \cdot \frac{u^{7/2}}{7/2} - \frac{u^{9/2}}{9/2} \right]_0^4 \\
&= \frac{1}{2} \left[\frac{8}{7} u^{7/2} - \frac{2}{9} u^{9/2} \right]_0^4 \\
&= \frac{1}{2} \left[\frac{8}{7} (u)^{7/2} - \frac{2}{9} (4)^{9/2} - 0 \right] \\
&= \frac{1}{2} \left[\frac{8}{7} (128) - \frac{2}{9} (512) \right] \\
&= \frac{1}{2} \left[\frac{1024}{7} - \frac{1024}{9} \right] \\
&= \frac{1024}{2} \cdot \frac{2}{63} = \frac{1024}{63} \\
&\therefore \int_0^2 x^8 \left(\frac{4}{x^2} - 1 \right)^{5/2} dx = \frac{1024}{63} = \frac{2^{10}}{63}
\end{aligned}$$

Question10

$$\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^2 x (\sin x + \cos x) dx =$$

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Options:

A.

0

B.

$\frac{2}{15}$

C.

$\frac{4}{15}$

D.

$$\frac{2}{5}$$

Answer: C

Solution:

$$\begin{aligned} & \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cdot \cos^2 x (\sin x + \cos x) dx \\ &= \int_{-\pi/2}^{\pi/2} \sin^3 x \cos^2 x dx \\ &+ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^3 x dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \cdot (1 - \cos^2 x) \cos^2 x dx \\ &+ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x (1 - \sin^2 x) \cdot \sin^2 x dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \cdot \cos^2 x \cdot dx \\ &- \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \cdot \cos^4 x dx \\ &+ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cdot \sin^2 x \cdot dx - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cdot \sin^4 x dx \\ &= \left[\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} \right]_{-\frac{\pi}{2}}^{\pi} \\ &= \left(0 + 0 + \frac{1}{3} - \frac{1}{5} \right) - \left(0 + 0 - \frac{1}{3} + \frac{1}{5} \right) \\ &= \frac{2}{3} - \frac{2}{5} = \frac{10 - 6}{15} = \frac{4}{15} \end{aligned}$$

Question11

$$\int_{1/5}^{1/2} \frac{\sqrt{x-x^2}}{x^3} dx =$$

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Options:

A.

$$\frac{21}{2}$$

B.

$$\frac{14}{3}$$

C.

$$\frac{7}{3}$$

D.

$$\frac{7}{2}$$

Answer: B

Solution:

$$\begin{aligned} & \int_{\frac{1}{5}}^{\frac{1}{2}} \frac{\sqrt{x-x^2}}{x^3} dx \\ &= \int_{\frac{1}{5}}^{\frac{1}{2}} \frac{x\sqrt{\frac{1}{x}-1}}{x^3} dx = \int_{\frac{1}{5}}^{\frac{1}{2}} \frac{1}{x^2} \sqrt{\frac{1}{x}-1} dx \end{aligned}$$

$$\text{Put } \frac{1}{x} - 1 = t \Rightarrow \frac{1}{x^2} dx = -dt$$

$$\text{at } x = \frac{1}{5} \Rightarrow t = 4 \text{ and at } x = \frac{1}{2} \Rightarrow t = 1$$

$$\begin{aligned} &= - \int_4^1 t^{\frac{1}{2}} \cdot dt = \int_1^4 \sqrt{t} \cdot dt \\ &= \left[\frac{t^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4 = \frac{2}{3} \left[t^{\frac{3}{2}} \right]_1^4 \\ &= \frac{2}{3} \left[4^{\frac{3}{2}} - 1^{\frac{3}{2}} \right] = \frac{2}{3} (8 - 1) \\ &= \frac{14}{3} \end{aligned}$$

Question12

$$\int_0^{400\pi} \sqrt{1 - \cos 2x} dx =$$

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Options:

A.

$$100\sqrt{2}$$

B.

$$200\sqrt{2}$$

C.

$$400\sqrt{2}$$

D.

$$800\sqrt{2}$$

Answer: D

Solution:

$$\begin{aligned} I &= \int_0^{400\pi} \sqrt{1 - \cos 2x} dx \\ &= \int_0^{400\pi} \sqrt{2 \sin^2 x} dx \\ &= \sqrt{2} \int_0^{400\pi} \sin x dx \\ &= 400\sqrt{2} \int_0^{\pi} \sin x dx \\ &= 2 \times 400\sqrt{2} \int_0^{\pi/2} \sin x dx \\ &= 800\sqrt{2} [-\cos x]_0^{\pi/2} \\ &= 800\sqrt{2} \left[-\cos \frac{\pi}{2} + \cos 0 \right] \\ &= 800\sqrt{2} \end{aligned}$$

Question13

$$\int_0^x \frac{t^2}{\sqrt{a^2+t^2}} dt =$$

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Options:

A.

$$\frac{x}{2}\sqrt{a^2+x^2} + \log \left| x + \sqrt{a^2+x^2} \right|$$

B.

$$\sqrt{a^2+x^2} - a^2 \sinh^{-1} \frac{x}{a}$$

C.

$$\frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{4} \log \left| x + \sqrt{a^2+x^2} \right|$$

D.

$$\frac{x}{2}\sqrt{a^2+x^2} - \frac{a^2}{2} \sinh^{-1} \frac{x}{a}$$

Answer: D

Solution:

$$\text{Let } I = \int_0^x \frac{t^2}{\sqrt{a^2+t^2}} dt$$

$$\text{Put } t = a \sinh \theta \Rightarrow dt = a \cosh \theta d\theta$$

$$\text{Now, } \sqrt{a^2+t^2} = \sqrt{a^2+a^2 \sinh^2 \theta}$$

$$= a \sqrt{1 + \sinh^2 \theta} = a \cosh \theta$$

$$\text{and } t^2 = a^2 \sinh^2 \theta$$

$$\begin{array}{c|c|c} t & 0 & x \\ \hline \theta & 0 & \sinh^{-1} \left(\frac{x}{a} \right) \end{array}$$

$$\therefore I = \int_0^{\sinh^{-1} \left(\frac{x}{a} \right)} \frac{a^2 \sinh^2 \theta}{a \cosh \theta} \cdot a \cosh \theta d\theta$$

$$= \int_0^{\sinh^{-1} \left(\frac{x}{a} \right)} a^2 \sinh^2 \theta d\theta$$

$$\therefore \sinh^2 \theta = \frac{\cosh(2\theta) - 1}{2}$$

$$\begin{aligned}
 \text{So, } I &= a^2 \int_0^{\sin h^{-1}(\frac{x}{a})} \frac{\cosh(2\theta) - 1}{2} d\theta = \frac{a^2}{2} \\
 &\left[\sin h^{-1} \left(\frac{x}{a} \right) \right. \\
 &\left. \int_0^{\cosh(2\theta)} d\theta - \int_0^{\sin h^{-1}(\frac{x}{a})} \int_1 1 \cdot d\theta \right] \\
 &= \frac{a^2}{2} \left[\frac{1}{2} \sinh(2\theta) - \theta \right] = \frac{a^2}{2} \\
 &= \frac{a^2}{2} \left[\frac{x}{a} \times \frac{\sqrt{a^2 + x^2}}{a} - \sinh^{-1} \left(\frac{x}{a} \right) \right] \\
 &= \frac{x}{2} \sqrt{a^2 + x^2} - \frac{a^2}{2} \sinh^{-1} \left(\frac{x}{a} \right)
 \end{aligned}$$

Question14

$$\int_{\frac{5}{6}}^{\pi} \cos^{-4} x dx =$$

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Options:

A.

$$\frac{64}{9\sqrt{3}}$$

B.

$$\frac{52\sqrt{3}}{9}$$

C.

$$\frac{62\sqrt{3}}{9}$$

D.

$$\frac{44}{9\sqrt{3}}$$

Answer: D

Solution:

$$\begin{aligned}
 \text{Let } I &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^4 x dx \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\cos^4 x} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^4 x dx \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x (1 + \tan^2 x) dx = \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} (1 + u^2) du
 \end{aligned}$$

x	$\frac{\pi}{6}$	$\frac{\pi}{3}$
u	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$

$$\begin{aligned}
 &\text{where } u = \tan x \\
 &= \left(u + \frac{u^3}{3} \right) \Big|_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \\
 &= \left(\sqrt{3} + \frac{(\sqrt{3})^3}{3} \right) - \left(\frac{1}{\sqrt{3}} + \frac{\left(\frac{1}{\sqrt{3}}\right)^3}{3} \right) \\
 &= \frac{3\sqrt{3} + 3\sqrt{3}}{3} - \left(\frac{1}{\sqrt{3}} + \frac{1}{9\sqrt{3}} \right) \\
 &= \frac{6\sqrt{3}}{3} - \left(\frac{9+1}{9\sqrt{3}} \right) = 2\sqrt{3} - \frac{10\sqrt{3}}{27} \\
 &= \frac{54\sqrt{3} - 10\sqrt{3}}{27} = \frac{44\sqrt{3}}{27} = \frac{44}{9\sqrt{3}}
 \end{aligned}$$

Question15

$$\int_0^{\frac{3\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx =$$

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Options:

A.

0

B.

1

C.

$$\frac{\pi}{4}$$

D.

$$\frac{3\pi}{4}$$

Answer: D

Solution:

$$\begin{aligned} \text{Let } I &= \int_0^{\frac{3\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx \quad \dots (i) \\ I &= \int_0^{\frac{3\pi}{2}} \frac{\left(\cos\left(\frac{3\pi}{2} - x\right)\right)^3}{\left(\cos\left(\frac{3\pi}{2} - x\right)\right)^3 + \left(\sin\left(\frac{3\pi}{2} - x\right)\right)^3} dx \\ &= \int_0^{\frac{3\pi}{2}} \frac{-\sin^3 x}{-\sin^3 x - \cos^3 x} dx \\ &= \int_0^{\frac{3\pi}{2}} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx \quad \dots (ii) \end{aligned}$$

On adding in Eqs. (i) and (ii), we get

$$\begin{aligned} \Rightarrow 2I &= \int_0^{\frac{3\pi}{2}} dx = \frac{3\pi}{2} \\ \therefore I &= \frac{3\pi}{4} \end{aligned}$$

Question16

If $k \in N$, then $\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{kn} \right] =$

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Options:

A.

$$\log(k+1)$$

B.

$$\log k$$

C.

$$\log(k+5)$$

D.

$$\log(k+1) - \log 6$$

Answer: B

Solution:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{kn} \right] \\ & \sum_{i=1}^{(k-1)n} \frac{1}{n+i} \\ & \Rightarrow \frac{1}{n} \sum_{i=1}^{(k-1)n} \frac{1}{1 + \frac{i}{n}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^{(k-1)n} \frac{1}{1 + \frac{i}{n}} \\ & = \int_0^{k-1} \frac{1}{1+x} dx = [\ln(1+x)]_0^{k-1} \\ & = \ln k \end{aligned}$$

Question17

$$\int_{-1}^4 \sqrt{\frac{4-x}{x+1}} dx =$$

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Options:

A.

$$0$$

B.

$$\frac{\pi}{2}$$

C.

$$\frac{3\pi}{2}$$

D.

$$\frac{5\pi}{2}$$

Answer: D

Solution:

$$\int_{-1}^4 \sqrt{\frac{4-x}{x+1}} dx$$

$$\text{Let } x = \frac{5}{2} \sin \theta + \frac{3}{2}$$

$$\Rightarrow dx = \frac{5}{2} \cos \theta d\theta$$

$$\Rightarrow 4 - x = 4 - \left(\frac{5}{2} \sin \theta + \frac{3}{2} \right)$$

$$= \frac{5}{2} (1 - \sin \theta)$$

$$\Rightarrow x + 1 = \frac{5}{2} \sin \theta + \frac{3}{2} + 1 = \frac{5}{2} (1 + \sin \theta)$$

$$\Rightarrow \frac{4-x}{x+1} = \frac{1-\sin \theta}{1+\sin \theta}$$

$$\Rightarrow \sqrt{\frac{4-x}{x+1}} = \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \frac{1-\sin \theta}{\cos \theta}$$

$$\int \sqrt{\frac{4-x}{x+1}} dx = \int \frac{1-\sin \theta}{\cos \theta} \frac{5}{2} \cos \theta d\theta$$

$$= \frac{5}{2} \int (1 - \sin \theta) d\theta$$

$$= \frac{5}{2} (\theta + \cos \theta) + c$$

$$x = \frac{5}{2} \sin \theta + \frac{3}{2} \Rightarrow \sin \theta = \frac{2x-3}{5}$$

$$\Rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta} = \frac{2}{5} \sqrt{4 + 3x - x^2}$$

$$\int_{-1}^4 \sqrt{\frac{4-x}{x+1}} dx = \left[\frac{5}{2} \sin^{-1} \left(\frac{2x-3}{5} \right) + \sqrt{4 + 3x - x^2} \right]_{-1}^4$$

$$= \frac{5}{2} \cdot \frac{\pi}{2} - \frac{5}{2} \left(\frac{-\pi}{2} \right)$$

$$= \frac{5\pi}{4} + \frac{5\pi}{4} = \frac{5\pi}{2}$$

Question 18

$$\int_0^{\pi/4} \frac{\cos^2 x}{\cos^2 x + 4 \sin^2 x} dx =$$

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Options:

A.

$$\frac{\pi}{2} - \frac{1}{3}\tan^{-1} 2$$

B.

$$-\frac{\pi}{4} - \frac{4}{3}\tan^{-1} 2$$

C.

$$\frac{\pi}{6} + \frac{2}{3}\tan^{-1} 2$$

D.

$$-\frac{\pi}{12} + \frac{2}{3}\tan^{-1} 2$$

Answer: D

Solution:

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} \frac{\cos^2 x dx}{\cos^2 x + 4 \sin^2 x} \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{1 + 4 \tan^2 x} dx \end{aligned}$$

Put $u = 2 \tan x$

$$du = 2 \sec^2 x dx$$

$$dx = \frac{du}{2(1 + \tan^2 x)} = \frac{2du}{4 + u^2}$$

$$x = 0, u = 0$$

$$x = \frac{\pi}{4}, u = 2$$

$$= \int_0^2 \frac{1}{1 + u^2} \cdot \frac{2du}{4 + u^2}$$

$$= \int_0^2 \left(\frac{2}{3} \frac{1}{1 + u^2} - \frac{2}{3} \frac{1}{4 + u^2} \right) du$$

$$= \frac{2}{3} [\tan^{-1} u]_0^2 - \frac{2}{3} \cdot \frac{1}{2} \left[\tan^{-1} \frac{u}{2} \right]_0^2$$

$$= \frac{2}{3} \tan^{-1}(2) - \frac{\pi}{12}$$

Question19

$$\int_{5\pi}^{25\pi} |\sin 2x + \cos 2x| dx =$$

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Options:

A.

$$20\sqrt{2}$$

B.

$$10\sqrt{2}$$

C.

$$40\sqrt{2}$$

D.

$$80\sqrt{2}$$

Answer: C

Solution:

$$\begin{aligned} & \int_{5\pi}^{25\pi} |\sin 2x + \cos 2x| dx \\ \Rightarrow & |\sin 2x + \cos 2x| = \sqrt{2} \left| \sin \left(2x + \frac{\pi}{4} \right) \right| \\ \Rightarrow & \int_{5\pi}^{25\pi} |\sin 2x + \cos 2x| dx \\ = & \int_{5\pi}^{25\pi} \sqrt{2} \left| \sin \left(2x + \frac{\pi}{4} \right) \right| dx \end{aligned}$$

$$\text{Let } u = 2x + \frac{\pi}{4}$$

$$du = 2dx$$

$$dx = \frac{1}{2}du$$

$$x = 5\pi, u = 10\pi + \frac{\pi}{4}$$

$$x = 25\pi, u = 50\pi + \frac{\pi}{4}$$

$$= \sqrt{2} \int_{10\pi + \frac{\pi}{4}}^{50\pi + \frac{\pi}{4}} |\sin u| \frac{1}{2} du$$

$$= \frac{\sqrt{2}}{2} \int_{10\pi + \frac{\pi}{4}}^{50\pi + \frac{\pi}{4}} |\sin u| du$$

The period of $|\sin u|$ is π .

The length of the interval of integration is

$$= \left(50\pi + \frac{\pi}{4}\right) - \left(10\pi + \frac{\pi}{4}\right) = 40\pi$$

$$= \frac{\sqrt{2}}{2} \cdot 40 \int_0^\pi |\sin u| du$$

$$= \frac{\sqrt{2}}{2} \times 40 \times 2$$

$$= 40\sqrt{2}$$

Question20

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{3}} \left| \tan \left(x - \frac{\pi}{6} \right) \right| dx =$$

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Options:

A.

$$\log \frac{\sqrt{3}-1}{\sqrt{6}}$$

B.

$$\log(2\sqrt{2}(\sqrt{3}+1))$$

C.

$$\log \frac{\sqrt{3}+1}{\sqrt{6}}$$

D.

$$\log(2\sqrt{2}(\sqrt{3}-1))$$

Answer: B

Solution:

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{3}} \left| \tan \left(x - \frac{\pi}{6} \right) \right| dx$$

$$\text{Put } \tan \left(x - \frac{\pi}{6} \right) = 0 \Rightarrow x - \frac{\pi}{6} = 0$$

$$\begin{aligned} I &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{6}} -\tan \left(x - \frac{\pi}{6} \right) dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan \left(x - \frac{\pi}{6} \right) dx \\ &= - \left[\log \left(\sec \left(x - \frac{\pi}{6} \right) \right) \right]_{-\frac{\pi}{4}}^{\frac{\pi}{6}} + \left[\log \sec \left(x - \frac{\pi}{6} \right) \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\ &\Rightarrow \log \left(\sec \left(\frac{5\pi}{12} \right) \right) + \log \left(\sec \left(\frac{\pi}{6} \right) \right) \\ &= \log \left(\sec \left(\frac{5\pi}{12} \right) \sec \left(\frac{\pi}{6} \right) \right) \\ &= \log \left(2\sqrt{2} \frac{(\sqrt{3}+1)}{\sqrt{3}} \right) \end{aligned}$$

Question21

$$\int_0^\pi \frac{x \sin x}{\sin^2 x + 2 \cos^2 x} dx =$$

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Options:

A.

$$\frac{\pi}{2}$$

B.

$$\frac{\pi^2}{2}$$

C.

$$\frac{\pi^2}{4}$$

D.

$$\frac{\pi}{4}$$

Answer: C

Solution:

$$\begin{aligned} I &= \int_0^\pi \frac{x \sin x}{\sin^2 x + 2 \cos^2 x} dx && \dots (i) \\ \Rightarrow I &= \int_0^\pi \frac{(\pi - x) \sin x}{\sin^2 x + 2 \cos^2 x} dx \\ &\quad \left[\because \int_0^a f(x) dx = \int_0^a f(a - x) dx \right] \\ \Rightarrow 2I &= \int_0^\pi \frac{\pi \sin x}{1 + \cos^2 x} dx \\ \text{Put } \cos x &= t \Rightarrow -\sin x dx = dt \\ \Rightarrow 2I &= -\pi \left[\tan^{-1} t \right]_1^{-1} \\ \Rightarrow 2I &= -\pi \left[\frac{-\pi}{4} - \frac{\pi}{4} \right] \Rightarrow I = \frac{\pi^2}{4} \end{aligned}$$

Question22

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1^2 + n^2} + \frac{2}{2^2 + n^2} + \frac{3}{3^2 + n^2} + \dots + \frac{n}{n^2 + n^2} \right) =$$

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Options:

A.

$$1$$

B.

$$\frac{1}{2} \log 2$$

C.

$$2 \log 2$$

D.

0

Answer: B

Solution:

$$\lim_{n \rightarrow \infty} \left[\frac{1}{1^2+n^2} + \frac{2}{2^2+n^2} + \frac{3}{3^2+n^2} + \frac{n}{n^2+n^2} \right]$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{r^2+n^2} = \lim_{n \rightarrow \infty} \sum \frac{\left(\frac{r}{n}\right)}{1+\left(\frac{r}{n}\right)^2} \cdot \frac{1}{n}$$

$$\Rightarrow \int_0^1 \frac{x}{1+x^2} dx$$

$$\text{put } 1+x^2 = t \Rightarrow 2x dx = dt$$

$$\Rightarrow \frac{1}{2} \int_1^2 \frac{1}{t} dt \Rightarrow \frac{1}{2} [\log t]_1^2 = \frac{1}{2} \log 2$$

Question23

$$\int_0^{\frac{\pi}{2}} \log |\tan x + \cot x| dx =$$

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Options:

A.

$$\pi \log 2$$

B.

$$-\pi \log 2$$

C.

$$\frac{\pi}{2} \log 2$$

D.

$$2\pi \log 2$$

Answer: A

Solution:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \ln(\tan x + \cot x) dx &= \int_0^{\frac{\pi}{2}} \ln \left(\frac{1}{\sin x \cos x} \right) dx \\&= \int_0^{\frac{\pi}{2}} (\ln 2 - \ln \sin 2x) dx \\&= \ln 2 \int_0^{\frac{\pi}{2}} dx - \int_0^{\frac{\pi}{2}} \ln \sin 2x dx \\&\text{put } 2x = t, dx = \frac{dt}{2} \\&= \frac{\pi}{2} \ln 2 - \frac{1}{2} \int_0^{\pi} \ln \sin t dt \\&= \frac{\pi}{2} \ln 2 - \int_0^{\pi/2} \ln \sin t dt \\&= \frac{\pi}{2} \ln 2 - \left(-\frac{\pi}{2} \ln 2 \right) \\&= \left[\because \int_0^{\pi/2} \ln \sin x = -\frac{\pi}{2} \ln 2 \right] \\&= \pi \ln 2\end{aligned}$$

Question24

$$\int_0^{\pi} x \cdot \sin^5 x \cdot \cos^6 x dx =$$

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Options:

A.

$$\frac{16\pi}{693}$$

B.

$$\frac{8\pi}{693}$$

C.

$$\frac{4\pi}{693}$$

D.

$$\frac{2\pi}{693}$$

Answer: B

Solution:

$$\text{Let } I = \int_0^\pi x \sin^5 x \cos^6 x dx$$

$$\text{Using } \int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx$$

$$\text{If } f(a+b-x) = f(x)$$

$$\begin{aligned} \Rightarrow I &= \frac{\pi}{2} \int_0^\pi \sin^5 x \cos^6 x dx \\ &= \frac{\pi}{2} 2 \int_0^{\pi/2} \sin^5 x \cos^6 x dx \end{aligned}$$

By Walli's Formula,

$$= \pi \left[\frac{(4 \cdot 2)(5 \cdot 3 \cdot 1)}{11 \cdot 9 \cdot 7 \cdot 5 \cdot 3 \cdot 1} \right] = \frac{8\pi}{693}$$

Question25

$$\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{1}{\left(x + \sqrt{1-x^2}\right)(1-x^2)} dx =$$

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Options:

A.

$$\log(\sqrt{3} + 1)$$

B.

$$\log(\sqrt{3} - 1)$$

C.

$$\log(3 + \sqrt{3})$$

D.

$$\log(3 - \sqrt{3})$$

Answer: D

Solution:

$$I = \int_{1/2}^{1/\sqrt{2}} \frac{1}{(x + \sqrt{1-x^2})(1-x^2)} dx$$

$$I = \int_{\pi/6}^{\pi/4} \frac{\cos \theta d\theta}{(\sin \theta + \cos \theta) \cos^2 \theta}$$

$$\text{Put } x = \sin \theta \Rightarrow dx = \cos \theta d\theta$$

$$= \int_{\pi/6}^{\pi/4} \frac{d\theta}{\sin \theta \cos \theta + \cos^2 \theta}$$

Dividing N^T and D^r by $\cos^2 \theta$, we get

$$= \int_{\pi/6}^{\pi/4} \frac{\sec^2 \theta}{\tan \theta + 1} d\theta$$

$$\text{Put } \tan \theta + 1 = t, \sec^2 \theta d\theta = dt$$

$$= \int_{\frac{1}{\sqrt{3}}+1}^2 \frac{dt}{t} = [\ln t]_{1+\frac{1}{\sqrt{3}}}^2$$

$$= \ln 2 - \ln \left(1 + \frac{1}{\sqrt{3}}\right)$$

$$= \ln 2 - \ln \left(\frac{\sqrt{3}+1}{\sqrt{3}}\right) = \ln \left(\frac{2\sqrt{3}}{\sqrt{3}+1}\right)$$

$$= \ln \left(\frac{2\sqrt{3}(\sqrt{3}-1)}{3-1}\right) = \ln(3 - \sqrt{3})$$

Question26

Let $H(x) = 3x^4 + 6x^3 - 2x^2 + 1$ and $g(x)$ be a linear polynomial. If

$$\frac{H(x)}{(x-1)(x+1)(x-2)} = f(x) + \frac{g(x)}{(x-1)(x+1)(x-2)}, \text{ then}$$

$$H(-1) + 2H(2) - 3H(1) =$$

Options:

A.

$$f(-1) + 2f(2) - 3f(1)$$

B.

$$H(-1) + f(2) + g(3)$$

C.

$$g(-1) + 2g(2) - 3g(1)$$

D.

$$H(1) + 2f(2) - g(1)$$

Answer: C

Solution:

$$\text{Given, } H(x) = 3x^4 + 6x^3 - 2x^2 + 1$$

$$\text{Given, } \frac{H(x)}{(x-1)(x-2)(x+1)}$$

$$= f(x) + \frac{g(x)}{(x-1)(x+1)(x-2)} \text{ taking LCM}$$

$$H(x) = f(x)(x-1)(x+1)(x-2) + g(x)$$

$$H(-1) = g(-1)$$

$$H(2) = g(2)$$

$$H(1) = g(1)$$

$$\text{Now, } H(-1) + 2H(2) - 3H(1)$$

$$= g(-1) + 2g(2) - 3g(1)$$

Question27

$$\int_{\pi/4}^{\pi/3} \frac{\cos x - \sin x}{\sin 2x} dx =$$

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Options:

A.

$$\frac{1}{2} \log \left[\frac{(3+2\sqrt{2})(2-\sqrt{3})}{\sqrt{3}} \right]$$

B.

$$\frac{1}{2} \log \left[\frac{(3-2\sqrt{2})(2+\sqrt{3})}{\sqrt{3}} \right]$$

C.

$$\log \left[\frac{(3-2\sqrt{2})(2-\sqrt{3})}{\sqrt{3}} \right]$$

D.

$$\log \left[\frac{(3+2\sqrt{2})(2-\sqrt{3})}{\sqrt{3}} \right]$$

Answer: A

Solution:

$$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos x - \sin x}{\sin 2x} dx$$

$$I = \int_{\sqrt{2}}^{\frac{\sqrt{3}}{2} + \frac{1}{2}} \frac{dt}{t^2 - 1}$$

$$\text{Put } \sin x + \cos x = t$$

$$(\cos x - \sin x)dx = dt$$

$$1 + \sin 2x = t^2$$

$$= \left[\frac{1}{2} \ln \left(\frac{t-1}{t+1} \right) \right]_{\sqrt{2}}^{\frac{\sqrt{3}+1}{2}}$$

$$= \frac{1}{2} \left[\ln \left(\frac{\frac{\sqrt{3}+1}{2} - 1}{\frac{\sqrt{3}+1}{2} + 1} \right) - \ln \left(\frac{\sqrt{2}-1}{\sqrt{2}+1} \right) \right]$$

$$= \frac{1}{2} \left[\ln \frac{\sqrt{3}-1}{\sqrt{3}+3} - \ln \left(\frac{\sqrt{2}-1}{\sqrt{2}+1} \right) \right]$$

$$= \frac{1}{2} \left[\ln \frac{2(2\sqrt{3}-3)}{6} - \ln(3-2\sqrt{2}) \right]$$

$$= \frac{1}{2} \left[\ln \frac{2\sqrt{3}-3}{3} + \ln(3+2\sqrt{2}) \right]$$

$$= \frac{1}{2} \left[\ln \left(\frac{2-\sqrt{3}}{\sqrt{3}} \right) + \ln(3+2\sqrt{2}) \right]$$

$$= \frac{1}{2} \ln \left[\frac{(3+2\sqrt{2})(2-\sqrt{3})}{\sqrt{3}} \right]$$

Question28

$$\int_0^{\pi/2} \frac{\sin x}{1+\cos x+\sin x} dx =$$

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Options:

A.

$$\frac{\pi}{2} + \frac{1}{2} \log 2$$

B.

$$\frac{\pi}{4} - \frac{1}{2} \log 2$$

C.

$$\frac{\pi}{4}$$

D.

$$\frac{3\pi}{4} + \log 2$$

Answer: B

Solution:

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \cos x + \sin x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin \frac{x}{2} + \cos \frac{x}{2} + \sin \frac{x}{2} - \cos \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(1 + \frac{\sin \frac{x}{2} - \cos \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \right) dx \\
&= \frac{1}{2} \left[x + \left\{ -2 \ln \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) \right\} \right]_0^{\frac{\pi}{2}} \\
&= \frac{1}{2} \left[x - 2 \ln \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) \right]_0^{\frac{\pi}{2}} \\
&= \frac{1}{2} \left[\left(\frac{\pi}{2} - 2 \ln \sqrt{2} \right) - 0 \right] \\
&= \frac{\pi}{4} - \ln \sqrt{2} = \frac{\pi}{4} - \frac{1}{2} \ln 2
\end{aligned}$$

Question29

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx =$$

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Options:

A.

$$\frac{\pi^2}{4}$$

B.

$$\frac{\pi}{2}$$

C.

$$\frac{\pi^2}{2}$$

D.

$$\frac{\pi}{4}$$

Answer: A

Solution:

$$I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \dots (i)$$

Apply property

$$\int_a^b xf(x)dx = \frac{a+b}{2} \int f(x)dx$$

$$\text{If } f(a+b-x) = f(x)$$

$$I = \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

$$\text{Put } \cos x = t$$

$$\sin x dx = -dt$$

$$I = -\frac{\pi}{2} \int_{+1}^{-1} \frac{dt}{1+t^2}$$

$$= \frac{\pi}{2} \int_{-1}^1 \frac{dt}{1+t^2} = \frac{\pi}{2} \cdot 2 \int_0^1 \frac{dt}{1+t^2}$$

$$= \pi (\tan^{-1} t)_0^1$$

$$= \pi \tan^{-1} 1 = \pi \times \frac{\pi}{4} = \frac{\pi^2}{4}$$

Question30

If $\int_0^{2\pi} (\sin^4 x + \cos^4 x) dx = K \int_0^\pi \sin^2 x dx + L \int_0^{\frac{\pi}{2}} \cos^2 x dx$ and $K, L \in N$, then the number of possible ordered pairs (K, L) is

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Options:

A. 1

B. 2

C. 3

D. 4

Answer: B

Solution:

$$\text{Let } I = \int_0^{2\pi} (\sin^4 x + \cos^4 x) dx$$

As $\sin^4 x + \cos^4 x$ is periodic with $\frac{\pi}{2}$

$$\begin{aligned}
& \text{So, } \int_0^{2\pi} (\sin^4 x + \cos^4 x) dx \\
&= \int_0^{4 \times \frac{\pi}{2}} (\sin^4 x + \cos^4 x) dx \\
&= 4 \int_0^{\pi/2} (\sin^4 x + \cos^4 x) dx \\
&= 4 \int_0^{\pi/2} 1 - \frac{\sin^2 2x}{2} dx \\
&= 4 \int_0^{\pi/2} \frac{4 - 1 + \cos 4x}{4} dx \\
&= 4 \int_0^{\pi/2} \frac{3 + \cos 4x}{4} dx \\
&= \left[3x - \frac{\sin 4x}{4} \right]_0^{\pi/2} \\
&= 3 \left[\frac{\pi}{2} \right] = \frac{3\pi}{2}
\end{aligned}$$

$$\begin{aligned}
& \text{Now, } \int_0^{\pi} \sin^2 x dx = 2 \int_0^{\frac{\pi}{2}} \sin^2 x dx \\
&= 2 \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos 2x}{2} \right) dx = \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2}
\end{aligned}$$

Similarly,

$$\begin{aligned}
& \int_0^{\frac{\pi}{2}} \cos^2 x dx = \frac{\pi}{4} \\
& \therefore \frac{3\pi}{2} = K \times \frac{\pi}{2} + L \times \frac{\pi}{4} \\
& 2K + L = 6
\end{aligned}$$

If $K = 1$, then $L = 4$ and if $K = 2$, then $L = 2$

$\therefore$ There are two possible ordered pair i.e. $(1, 4), (2, 2)$.

Question 31

$$\int_0^{\pi} \frac{x \sin x}{4 \cos^2 x + 3 \sin^2 x} dx \text{ is equal to}$$

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Options:

A. $\frac{\pi^2}{6\sqrt{3}}$

B. $\frac{\pi}{3\sqrt{3}}$

C. $\frac{\pi^2}{3\sqrt{3}}$

D. $\sqrt{3}\pi^2$

Answer: A

Solution:

$$\begin{aligned} \text{Let } I &= \int_0^\pi \frac{x \sin x dx}{4 \cos^2 x + 3 \sin^2 x} \quad \dots (i) \\ &= \int_0^\pi \frac{x \sin x dx}{4 \cos^2 x + 3 - 3 \cos^2 x} \\ &= \int_0^\pi \frac{x \sin x}{\cos^2 x + 3} \\ &= \int_0^\pi \frac{(\pi - x) \sin(\pi - x)}{\cos^2(\pi - x) + 3} \end{aligned}$$

$$= \int_0^\pi \frac{(\pi - x) \sin x}{\cos^2 x + 3} \quad \dots (ii)$$

On adding Eq. (i) and (ii), we get

$$\begin{aligned} 2I &= \int_0^\pi \frac{\pi \sin x}{3 + \cos^2 x} dx \\ &= \pi \int_0^\pi \frac{\sin x}{3 + \cos^2 x} dx = -\pi \int_1^{-1} \frac{1}{3 + t^2} dt \\ &= -\pi \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{t}{\sqrt{3}} \right)_1^{-1} \\ \Rightarrow 2I &= \frac{-\pi}{\sqrt{3}} \left[\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) - \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right] \\ &= \frac{-\pi}{\sqrt{3}} \left[-\frac{\pi}{6} - \frac{\pi}{6} \right] \\ &= -\frac{\pi}{\sqrt{3}} \times \left(-\frac{2\pi}{6} \right) = \frac{\pi^2}{3\sqrt{3}} \\ \Rightarrow I &= \frac{\pi^2}{6\sqrt{3}} \end{aligned}$$

Question32

If $A = \int_0^\infty \frac{1+x^2}{1+x^4} dx$, $B = \int_0^1 \frac{1+x^2}{1+x^4} dx$, then

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Options:

A. $2A = B$

B. $A = B$

C. $2B = A$

D. $2B + A = 0$

Answer: C

Solution:

$$\begin{aligned}\text{Let } I &= \int \left(\frac{1+x^2}{1+x^4} \right) dx \\ &= \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x^2 + \frac{1}{x^2} - 2 + 2\right)} dx \\ &= \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2} dx\end{aligned}$$

$$\text{Let } x - \frac{1}{x} = t$$

$$\begin{aligned}\Rightarrow \left(1 + \frac{1}{x^2}\right) dx &= dt \\ &= \int \frac{1}{t^2 + 2} dt = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 1}{\sqrt{2}x} \right) \text{ or } \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x - \frac{1}{x}}{\sqrt{2}} \right)\end{aligned}$$

$$\begin{aligned}A &= \int_0^\infty \frac{1+x^2}{1+x^4} dx = \left[\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x - \frac{1}{x}}{\sqrt{2}} \right) \right]_0^\infty \\ &= \frac{1}{\sqrt{2}} [\tan^{-1} \infty - \tan^{-1}(-\infty)] \\ &= \frac{1}{\sqrt{2}} \times 2 \times \frac{\pi}{2} \Rightarrow \frac{\pi}{\sqrt{2}}\end{aligned}$$

$$\begin{aligned}B &= \int_0^1 \frac{1+x^2}{1+x^4} dx = \left[\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x - \frac{1}{x}}{\sqrt{2}} \right) \right]_0^1 \\ &= \frac{1}{\sqrt{2}} [\tan^{-1}(0) - \tan^{-1}(-\infty)] = \frac{1}{\sqrt{2}} \times \frac{\pi}{2} \\ &\Rightarrow A = 2B\end{aligned}$$

Question33

$\int_0^1 \sqrt{\frac{2+x}{2-x}} dx$ is equal to

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Options:

A. $\pi + 2$

B. $\frac{1}{2}(\pi + 2)$

C. $\frac{\pi}{2} + 2 + \sqrt{3}$

D. $\frac{\pi}{3} + 2 - \sqrt{3}$

Answer: D

Solution:

We have, $I = \int_0^1 \sqrt{\frac{2+x}{2-x}} dx$

Let $\frac{2+x}{2-x} = u$

$$\left(\frac{1}{2-x} + \frac{x+2}{(2-x)^2} \right) dx = du$$

Now, lower limit = 1

and upper limit = 3

$$= \int_1^3 \frac{\sqrt{u}}{(u+1)^2} du$$

Now, let $s = \sqrt{u}$

$$\begin{aligned} ds &= \frac{1}{2\sqrt{u}} \cdot du = 8 \int_1^{\sqrt{3}} \frac{s^2}{(s^2+1)^2} ds \\ &= 8 \int_1^{\sqrt{3}} \left(\frac{1}{s^2+1} - \frac{1}{(s^2+1)^2} \right) ds \\ &= 8 \left| \tan^{-1}(s) \right|_1^{\sqrt{3}} - 8 \int_1^{\sqrt{3}} \frac{1}{(1+s^2)^2} ds \\ &= 8 \left(\tan^{-1}(\sqrt{3}) - \tan^{-1}(1) \right) - 8 \end{aligned}$$

$$\int_1^{\sqrt{3}} \frac{1}{(1+s^2)^2} ds$$

$$\begin{aligned}
&= 8 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) - 8 \int_1^{\sqrt{3}} \frac{1}{(1+s^2)^2} ds \\
&= \frac{2\pi}{3} - 8 \int_{\pi/4}^{\pi/3} \cos^2 p dp
\end{aligned}$$

$$[\text{put } s = \tan p]$$

$$\begin{aligned}
&= \frac{2\pi}{3} + 2 - \sqrt{3} + 1 - 4p \Big|_{\pi/4}^{\pi/3} \\
&= \frac{1}{3}(6 - 3\sqrt{3} + \pi) = \frac{\pi}{3} + 2 - \sqrt{3}
\end{aligned}$$

Question34

If $M = \int_0^{\infty} \frac{\log t}{1+t^3} dt$ and $N = \int_{-\infty}^{\infty} \frac{te^{2t}}{1+e^{3t}} dt$, then

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Options:

A. $N = 2M$

B. $N = M$

C. $N = 3M$

D. $N = -M$

Answer: D

Solution:

We have,

$$M = \int_0^{\infty} \frac{\log t}{1+t^3} dt \text{ and } N = \int_{-\infty}^{\infty} \frac{te^{2t}}{1+e^{3t}} dt$$

$$N = \int_{-\infty}^{\infty} \frac{t \cdot e^{2t}}{1+e^{3t}} dt$$

$$\text{Let } e^t = x$$

$$e^t \cdot dt = dx$$

$$\text{Lower limit} = 0$$

$$\text{and Upper limit} = \infty$$

$$\begin{aligned}
 N &= \int_0^\infty \frac{\log x \cdot x^2}{1+x^3} \cdot \frac{dx}{x} = \int_0^\infty \frac{x \log x}{1+x^3} dx \\
 &= \int_0^\infty \frac{x \log x}{1+x^3} dx = - \int_0^\infty \frac{\log x}{1+x^3} dx \\
 \Rightarrow N &= - \int_0^\infty \frac{\log t}{1+t^3} dt \Rightarrow N = -M
 \end{aligned}$$

Question 35

$\int_{-2}^2 (4 - x^2)^{\frac{5}{2}} dx$ is equal to

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Options:

A. 40π

B. 20π

C. 10π

D. $\frac{5\pi}{32}$

Answer: B

Solution:

To solve the integral $I = \int_{-2}^2 (4 - x^2)^{\frac{5}{2}} dx$, we can use a substitution method.

Substitution:

Substitute $x = 2 \sin u$:

Then $dx = 2 \cos u \, du$.

Change the limits:

When $x = -2$, $u = -\frac{\pi}{2}$.

When $x = 2$, $u = \frac{\pi}{2}$.

Substitute into the integral:

$$I = \int_{-\pi/2}^{\pi/2} (4 - (2 \sin u)^2)^{\frac{5}{2}} \cdot 2 \cos u \, du$$

$$= \int_{-\pi/2}^{\pi/2} (4 - 4 \sin^2 u)^{\frac{5}{2}} \cdot 2 \cos u \, du$$

Simplify:

Since $4 - 4 \sin^2 u = 4 \cos^2 u$, we have:

$$= \int_{-\pi/2}^{\pi/2} (4 \cos^2 u)^{\frac{5}{2}} \cdot 2 \cos u \, du$$

$$= 32 \int_{-\pi/2}^{\pi/2} \cos^6 u \, du$$

Using Reduction Formula:

Solving this integral using the reduction formula yields:

$$I = 64 \int_{-\pi/2}^{\pi/2} \cos^6 u \, du = 20\pi$$

Thus, the value of the integral is $I = 20\pi$.

Question 36

$\lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{n^3}\right)^{\frac{1}{n^3}} \left(1 + \frac{8}{n^3}\right)^{\frac{4}{n^3}} \left(1 + \frac{27}{n^3}\right)^{\frac{9}{n^3}} \dots (2)^{\frac{1}{n}} \right]$ is equal to

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Options:

A. $\log 2 - \frac{1}{2}$

B. $e^{(\log 2 - \frac{1}{2})}$

C. $e^{\left(\frac{2 \log 2 - 1}{3}\right)}$

D. $\frac{1}{3}(2 \log 2 - 1)$

Answer: C

Solution:

We have, $P = \lim_{x \rightarrow \infty}$

$$\left[\left(1 + \frac{1}{n^3}\right)^{1/n^3} - \left(1 + \frac{8}{n^3}\right)^{4/n^3} \dots (2)^{1/n} \right]$$

$$\Rightarrow \log p = \lim_{x \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{r^2}{n^2} \log \left(1 + \frac{r^3}{n^3}\right)$$

$$\Rightarrow \log p = \int_0^1 x^2 \log(1 + x^3) dx$$

$$\text{Put } 1 + x^3 = t \Rightarrow 3x^2 dx = dt$$

$$\text{when } n = 0, t = 1 \text{ and } x = 1, t = 2$$

$$\therefore \log p = \frac{1}{3} \int_1^2 \log t dt = \frac{1}{3} [t \log t - t]_1^2$$

$$\Rightarrow \log p = \frac{1}{3} [2 \log 2 - 1]$$

$$\text{Hence, } p = e^{(2 \log 2 - 1/3)}$$

Question 37

$$\int_{-5\pi}^{5\pi} (1 - \cos 2x)^{\frac{5}{2}} dx \text{ is equal to}$$

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Options:

A. $\frac{64\sqrt{2}}{5}$

B. $\frac{128\sqrt{2}}{5}$

C. $\frac{256\sqrt{2}}{3}$

D. $\frac{128\sqrt{2}}{3}$

Answer: D

Solution:

To solve the integral $\int_{-5\pi}^{5\pi} (1 - \cos 2x)^{\frac{5}{2}} dx$, we start by simplifying the expression inside the integral:

Notice that $1 - \cos 2x = 2 \sin^2 x$. Substituting this in, the integral becomes:

$$I = \int_{-5\pi}^{5\pi} (1 - \cos 2x)^{\frac{5}{2}} dx = \int_{-5\pi}^{5\pi} (2 \sin^2 x)^{\frac{5}{2}} dx$$

Simplify the expression:

$$= \int_{-5\pi}^{5\pi} 2^{\frac{5}{2}} \cdot |\sin^5 x| dx$$

Since $|\sin^5 x| = \sin^5 x$ for the interval $[0, \pi]$ and is symmetrical over $[-\pi, 0]$, the integral becomes:

$$= 8\sqrt{2} \int_0^{5\pi} \sin^5 x dx$$

Exploit the periodicity of the sine function to break the integral over one full period:

$$= 40\sqrt{2} \int_0^{\pi} \sin^5 x dx$$

Using the reduction formula for $\int_0^{\pi} \sin^n x dx$, the final result of this reduction gives:

$$= \frac{128\sqrt{2}}{3}$$

Question38

$$\int_0^{\pi/4} \log(1 + \tan x) dx =$$

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Options:

A. $\pi \log 2 + 1$

B. $\frac{\pi}{2} \log 2 + 1$

C. $\frac{\pi}{4} \log 2$

D. $\frac{\pi}{8} \log 2$

Answer: D

Solution:

$$I = \int_0^{\pi/4} \log(1 + \tan x) dx \quad \dots (i)$$

$$\text{Using } \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\begin{aligned}
 I &= \int_0^{\pi/4} \log \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx \\
 &= \int_0^{\pi/4} \log \left(1 + \frac{1 - \tan x}{1 + \tan x} \right) dx \\
 &\quad \left(\because \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} \right) \\
 I &= \int_0^{\pi/4} \log \left(\frac{2}{1 + \tan x} \right) dx \quad \dots (ii)
 \end{aligned}$$

On adding Eq. (i) and Eq. (ii), we get

$$\begin{aligned}
 2I &= \int_0^{\pi/4} \log(1 + \tan x) dx \\
 &\quad + \int_0^{\pi/4} \log \frac{2}{(1 + \tan x)} dx \\
 &= \int_0^{\pi/4} \log(1 + \tan x) \cdot \frac{2}{(1 + \tan x)} dx \\
 &= \int_0^{\pi/4} \log 2 dx = \log 2 \int_0^{\pi/4} dx \\
 &= \log 2 \cdot (x)_0^{\pi/4} = \log 2 \cdot \left(\frac{\pi}{4} - 0 \right) \\
 2I &= \frac{\pi}{4} \log 2 \Rightarrow I = \frac{\pi}{8} \log 2
 \end{aligned}$$

Question39

$$\int_{\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx =$$

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Options:

- A. $\frac{3\pi^2}{4}$
- B. $\frac{\pi}{2} + 1$
- C. $\frac{\pi^2}{4}$
- D. $\frac{\pi^2}{2}$

Answer: D

Solution:

$$\begin{aligned} I &= \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\ &= 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\ &= 2 \left[\int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \right] \\ &= 2 \left[\pi \int_0^{\pi} \frac{\sin x dx}{1 + \cos^2 x} - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \right] \\ 2I &= 2\pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \end{aligned}$$

Let $t = \cos x$

$$dt = -\sin x dx$$

$$= -\pi \int_1^{-1} \frac{1}{1 + t^2} dt = \pi \int_{-1}^1 \frac{1}{1 + t^2} dt$$

$$\begin{aligned} I &= \pi [\tan^{-1}(t)]_{-1}^1 \\ &= \pi \left[\frac{\pi}{4} + \frac{\pi}{4} \right] = \frac{\pi^2}{2} \end{aligned}$$

Question40

$$\int_0^{\pi/4} \frac{x^2}{(x \sin x + \cos x)^2} dx =$$

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Options:

A. $\frac{2-\pi}{2+\pi}$

B. $\frac{4-\pi}{4+\pi}$

C. $\frac{6-\pi}{6+\pi}$

D. $\frac{8-\pi}{8+\pi}$

Answer: B

Solution:

We have,

$$I = \int_0^{\pi/4} \frac{x^2 dx}{(x \sin x + \cos x)} \Rightarrow I = \int_0^{\pi/4} \frac{x \sec x \cdot \cos x \cdot dx}{(x \sin x + \cos x)^2}$$

$$\Rightarrow I = \int_0^{\pi/4} x \sec x \cdot \frac{x \cos x}{(x \sin x + \cos x)^2} dx$$

Let $I_1 = x \sec x$ and $I_2 = \frac{x \cos x}{x \sin x + \cos x}$

Let $x \sin x + \cos x = t$

$$x \cos x + \sin x - \sin x = \frac{dt}{dx}$$

$$x \cos x dx = dt$$

$$\Rightarrow I_2 = \int \frac{x \cos x}{(x \sin x + \cos x)^2} = \int \frac{dt}{t^2} = -\frac{1}{t}$$

$$\Rightarrow I_2 = \frac{-1}{(x \sin x + \cos x)}$$

$$\Rightarrow I = \int_0^{\pi/4} x \sec x \cdot \frac{x \cos x}{(x \sin x + \cos x)} dx$$

$$\Rightarrow I = \left[x \sec x \cdot \frac{-1}{(x \sin x + \cos x)} \right]_0^{\pi/4} - \int_0^{\pi/4} [\sec x + x \sec x \tan x] \cdot \frac{-1}{(x \sin x + \cos x)} dx$$

$$\therefore \int f(x)g(x)dx = f(x) \int g(x)dx - \int \left[\frac{d}{dx} f(x) - \int g(x)dx \right] dx$$

$$\Rightarrow I = \left[\frac{-x \sec x}{x \sin x + \cos x} \right]_0^{\pi/4} + \int_0^{\pi/4} \frac{\sec x(1+x \tan x)dx}{(x \sin x + \cos x)}$$

$$\Rightarrow I = \left[\frac{-x \sec x}{x \sin x + \cos x} \right]_0^{\pi/4} + \int_0^{\pi/4} \frac{\sec^2 x(\cos x + x \sin x)}{(x \sin x + \cos x)} dx$$

$$\Rightarrow I = \left[\frac{-x \sec x}{(x \sin x + \cos x)} \right]_0^{\pi/4} + \int_0^{\pi/4} \sec^2 x dx$$

$$\Rightarrow I = \frac{-\frac{\pi}{4} \cdot \sqrt{2}}{\frac{\pi}{4} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}} + \int_0^{\pi/4} \sec^2 x dx$$

$$\Rightarrow I = \frac{-\pi\sqrt{2}/4}{(\pi+4)/4\sqrt{2}} + [\tan x]_0^{\pi/4}$$

$$\Rightarrow I = \frac{-2\pi}{\pi+4} + (1-0) = \frac{-2\pi}{\pi+4} + 1 \Rightarrow I = \frac{4-\pi}{\pi+4}$$

Question41

$$\int_0^1 \frac{x}{(1-x)^{\frac{3}{4}}} dx =$$

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Options:

A. $\frac{4}{5}$

B. $\frac{8}{15}$

C. $\frac{14}{5}$

D. $\frac{16}{5}$

Answer: D

Solution:

We have, $I = \int_0^1 \frac{x}{(1-x)^{\frac{3}{4}}} dx$

Let $1 - x = t \Rightarrow x = 1 - t \Rightarrow -dx = dt$

$$\therefore I = \int_0^1 \frac{1-t}{t^{3/4}} dt \Rightarrow I = \int_0^1 t^{-3/4}(1-t) dt$$

$$\Rightarrow I = \int_0^1 t^{-3/4} dt - \int_0^1 t^{1/4} dt \Rightarrow I = \left[4t^{1/4} \right]_0^1 - \left[\frac{4t^{5/4}}{5} \right]_0^1$$

Putting the value of t

$$I = [4(1-x)]_0^1 + \left[\frac{4}{5}(1-x)^{5/4} \right]_0^1$$

On solving, we get

$$I = \frac{16}{5}$$

Question42

$$\int_{-1}^1 \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2} \right) dx =$$

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Options:

A. 2

B. 4

C. 0

D. 8

Answer: C

Solution:

We have,

$$I = \int_{-1}^1 \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2} \right) dx$$

$$\text{Let } f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$$

$$\Rightarrow f(-x) = \sqrt{1-x+x^2} - \sqrt{1+x+x^2}$$

$$\Rightarrow f(-x) = - \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2} \right) \Rightarrow f(-x) = -f(x)$$

$\therefore f(x)$ is an odd function.

As, we know that

$$\int_{-a}^a f(x) dx = \begin{cases} \int_0^a f(x) dx, & \text{if } f(-x) = f(x), \text{ even function} \\ 0, & \text{if } f(-x) = -f(x), \text{ odd function} \end{cases}$$

$$\text{Then, } I = \int_{-1}^1 \sqrt{1+x+x^2} - \sqrt{1-x+x^2} dx = 0$$

Question43

$$\int_1^5 (|x-3| + |1-x|) dx =$$

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Options:

A. 4

B. 8

C. 12

D. 24

Answer: C

Solution:

We have,

$$\Rightarrow I = \int_1^5 |x-3| + |1-x| dx$$

Let $f(x) = |x - 3| + |1 - x|$

Here, when $1 \leq x \leq 3$

$$\Rightarrow f(x) = -x - 3 - 1 + x = 2$$

When, $3 \leq x \leq 5$

$$\Rightarrow f(x) = x - 3 - 1 + x = 2x - 4$$

$$\begin{aligned}\Rightarrow I &= \int_1^5 |x - 3| + |1 - x| dx \\ &= \int_1^3 2dx + \int_3^5 (2x - 4)dx\end{aligned}$$

$$\Rightarrow I = [2x]_1^3 + \left[\frac{2x^2}{2} - 4x \right]_3^5$$

On solving, we get

$$I = 12$$

Question44

If $729 \int_1^3 \frac{1}{x^3(x^2+9)^2} dx = a + \log b$, then $(a - b) =$

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Options:

A. 4

B. $-\frac{4}{5}$

C. $\frac{4}{5}$

D. -4

Answer: A

Solution:

If $729 \int_1^3 \frac{1}{x^3(x^2+9)^2} dx = a + \log b$ Then, $a - b$

On substituting $u = x^2$

$$\Rightarrow du = 2x dx$$

$$\Rightarrow dx = \frac{1}{2} \frac{du}{\sqrt{u}}$$

$$\text{when, } x = 1$$

$$\Rightarrow u = 1$$

$$x = 3$$

$$\Rightarrow u = 9$$

$$\begin{aligned} I &= 729 \int_1^9 \frac{1}{u^{\frac{3}{2}}(u+9)^2} du \\ &= 729 \int_1^9 \frac{1}{2u^{\frac{3}{2}+\frac{1}{2}}(u+9)^2} du \\ &= 729 \int_1^9 \frac{1}{2u^2(u+9)^2} du \end{aligned}$$

Partial fraction

$$\frac{1}{u^2(u+9)^2} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{(u+9)} + \frac{D}{(u+9)}$$

$$\frac{1}{u^2(u+9)^2} \frac{A(u(u+9)^2 + B(u+9)^2 + Cu^2(u+9) + Du^2)}{u^2(u+9)^2}$$

$$1 = Au(u^2 + 81 + 18u) + B(u^2 + 81 + 18u) + Cu^3 + 9Cu^2 + Du^2$$

$$1 = Au^3 + 81Au + 18Au^2 + Bu^2 + 81B + 18uB + Cu^3 + 9Cu^2 + Du^2$$

$$1 = (A+C)u^3 + (18A+B+9C+D)u^2 + (81A+18B)u + 81B$$

Comparison of coefficient u^3, u^2, u and 1.

$$A+C=0, 18A+B+9C+D=0$$

$$81A+18B=0 \Rightarrow 81B=1$$

$$B = \frac{1}{81} \Rightarrow 81A + 18 \cdot \frac{1}{81} = 0$$

$$81A = -\frac{2}{9} \Rightarrow A = \frac{-2}{729}$$

$$\frac{-2}{729} + C = 0 \Rightarrow C = \frac{2}{729}$$

$$18\left(\frac{-2}{729}\right) + \frac{1}{81} + 9\left(\frac{2}{729}\right) + D = 0$$

$$-\frac{4}{81} + \frac{1}{81} + \frac{2}{81} + D = 0 \Rightarrow \frac{-1}{81} + D = 0$$

$$\begin{aligned}
D &= \frac{1}{81} \\
&= \frac{729}{2} \int_1^9 \left(\left(\frac{-2}{729} \right) \frac{1}{u} + \frac{1}{81u^2} + \frac{2}{729(u+9)} + \frac{1}{81(u+9)^2} \right) du \\
&= \frac{729}{2} \left[\frac{-2}{729} \log u + \frac{1}{81} \frac{u^{-2+1}}{(-2+1)} + \frac{2}{729} \log(u+9) + \frac{1}{81} \frac{(u+9)^{-2+1}}{-2+1} \right]_1^9 \\
&= \frac{729}{2} \left[\frac{-2}{729} (\log 9 - \log 1) + \frac{1}{81} (-1) \left(\frac{1}{9} - 1 \right) + \frac{2}{729} [\log(18) - \log(10)] \right] \\
&= \frac{729}{2} \left[\frac{-2}{729} \log 9 - \frac{1}{81} \left(\frac{-8}{9} \right) + \frac{2}{729} \log \left(\frac{18}{10} \right) - \frac{1}{81} \left(\frac{-8}{180} \right) \right] + C \\
&= \frac{1}{2} \left[-2 \log 9 + 8 + 2 \log \left(\frac{1}{5} \right) + \frac{8}{20} \right] + C \\
&= \frac{1}{2} \left[-2 \log 9 + 8 + 2 \log 9 - 2 \log 5 + \frac{2}{5} \right] + C \\
&= \frac{1}{2} \left[8 + \frac{2}{5} - 2 \log 5 \right] + C \\
&= \frac{1}{2} \left[\frac{40+2}{5} - 2 \log 5 \right] \Rightarrow \frac{42}{2 \times 5} - \log 5
\end{aligned}$$

$$a = \frac{42}{2 \times 5}, b = \frac{1}{5}$$

$$a = \frac{21}{5}, b = \frac{1}{5}$$

$$a - b = \frac{21}{5} - \frac{1}{5} = \frac{20}{5} = 4$$

Question45

$$\lim_{n \rightarrow \infty} \frac{1^{17} + 2^{77} + \dots + n^{77}}{n^{78}} =$$

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Options:

A. $\frac{1}{77}$

B. 1

C. 76

D. $\frac{1}{78}$

Answer: D

Solution:

To find the limit

$$\lim_{n \rightarrow \infty} \frac{1^{77} + 2^{77} + \dots + n^{77}}{n^{78}}$$

we use the method of integral approximation to approximate the sum of powers. Consider the sum

$$S_n = 1^{77} + 2^{77} + \dots + n^{77}.$$

This can be approximated using the integral

$$S_n \approx \int_1^n x^{77} dx.$$

Evaluating this integral, we have

$$\int_1^n x^{77} dx = \left[\frac{x^{78}}{78} \right]_1^n = \frac{n^{78}}{78} - \frac{1}{78}.$$

As n becomes very large, the term $\frac{1}{78}$ becomes negligible, and we approximate

$$S_n \approx \frac{n^{78}}{78}.$$

Thus, the given limit can be rewritten and evaluated as follows:

$$\lim_{n \rightarrow \infty} \frac{S_n}{n^{78}} = \lim_{n \rightarrow \infty} \frac{\frac{n^{78}}{78}}{n^{78}} = \lim_{n \rightarrow \infty} \frac{1}{78} = \frac{1}{78}.$$

Therefore, the value of the limit is

$$\frac{1}{78}.$$

Question46

$$\text{If } f(x) = \begin{cases} \frac{6x^2+1}{4x^3+2x+3} & , 0 < x < 1 \\ x^2 + 1 & , 1 \leq x < 2 \end{cases}, \text{ then } \int_0^2 f(x) dx =$$

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Options:

A. $\frac{1}{2} \log 3 + \frac{10}{3}$

B. $\frac{1}{2} \log 3 - \frac{10}{3}$

C. $\frac{1}{2} \log 3 + \frac{13}{3}$

D. $\frac{1}{2} \log 3 + \frac{20}{3}$

Answer: A

Solution:

$$f(x) = \begin{cases} \frac{6x^2+1}{4x^3+2x+3}, & 0 < x < 1 \\ x^2 + 1 & 1 \leq x < 2 \end{cases}$$

$$\int_0^2 f(x)dx = \int_0^1 \left(\frac{6x^2+1}{4x^3+2x+3} \right) dx + \int_1^2 (x^2+1)dx$$

$$\text{Let } u = 4x^3 + 2x + 3$$

$$du = (12x^2 + 2)dx \Rightarrow 2(6x^2 + 1)dx$$

$$\Rightarrow dx = \frac{du}{2(6x^2 + 1)}$$

$$x = 0 \Rightarrow u = 3$$

$$x = 1 \Rightarrow u = 4 + 2 + 3 = 9$$

$$\text{Or } (6x^2 + 1)dx = \frac{du}{2} \Rightarrow \int_3^9 \frac{1}{u} \left(\frac{du}{2} \right) + \int_1^2 (x^2 + 1)dx$$

$$\frac{1}{2} [\log u]_3^9 + \left[\frac{x^3}{3} + x \right]_1^2$$

$$\frac{1}{2} [\log 9 - \log 3] + \left[\frac{8}{3} + 2 - \frac{1}{3} - 1 \right]$$

$$\frac{1}{2} \left[\log \left(\frac{9}{3} \right) + \left(\frac{8+6}{3} - \frac{4}{3} \right) \right]$$

$$\frac{1}{2} \log 3 + \frac{14}{3} - \frac{4}{3} = \frac{1}{2} \log 3 + \frac{10}{3}$$

Question47

If $\int_1^n [x]dx = 120$, then $n =$

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Options:

A. 15

B. 16

C. 14

D. 12

Answer: B

Solution:

We have,

$$\begin{aligned}\int_1^n [x] dx &= 120 \\&= \int_1^2 1 dx + \int_2^3 2 dx + \dots + \int_{n-1}^n (n-1) dx = 120 \\&\Rightarrow [x]_1^2 + [2x]_2^3 + \dots + [(n-1)x]_{n-1}^n = 120 \\&\Rightarrow 2 - 1 + 6 - 4 + \dots + (n-1) = 120 \\&\Rightarrow 1 + 2 + \dots + n - 1 = 120 \\&\Rightarrow \frac{n(n-1)}{2} = 120 \\&\Rightarrow n^2 - n - 240 = 0 \\&\Rightarrow n^2 - 16n + 15n - 240 = 0 \\&\Rightarrow n(n-16) + 15(n-16) = 0 \\&(n-16)(n+15) = 0 \\&n = 16\end{aligned}$$

Question48

$$\int_{-\frac{1}{24}}^{\frac{1}{24}} \sec x \log \left(\frac{1-x}{1+x} \right) dx =$$

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Options:

A. $\frac{\pi}{2}$

B. π

C. 1

D. 0

Answer: D

Solution:

$$\int_{-1/24}^{1/24} \sec x \log \left(\frac{1-x}{1+x} \right) dx$$

Here, $f(x) = \sec x \log \left(\frac{1-x}{1+x} \right)$

Now, $f(-x) = \sec(-x) \log \left(\frac{1-(-x)}{1-(-x)} \right)$

$$= \sec x \log \left(\frac{1+x}{1-x} \right)$$

$$= -\sec x \log \left(\frac{1-x}{1+x} \right) = -f(x)$$

Thus, the given function is an odd function

$$\therefore \int_{-\frac{1}{24}}^{\frac{1}{24}} \sec x \log \left(\frac{1-x}{1+x} \right) dx = 0$$

$$\left[\because \int_{-a}^a f(x) dx = 0, \text{ if } f(x) \text{ is an odd function} \right]$$

Question49

If $[x]$ is the greatest integer function, then $\int_0^5 [x] dx =$

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Options:

A. 15

B. 2

C. 3

D. 10

Answer: D

Solution:

$$I = \int_0^5 [x] dx$$

$$= \int_0^1 0 dx + \int_1^2 1 dx + \int_2^3 2 dx + \int_3^4 3 dx + \int_4^5 4 dx$$

$$= 0 + (2-1) + 2(3-2) + 3(4-3) + 4(5-4) = 1 + 2 + 3 + 4 = 10$$

Question50

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\sqrt{\tan x}} dx =$$

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Options:

A. 0

B. $\frac{\pi}{2}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{4}$

Answer: D

Solution:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}} dx \quad \dots (i) \\ &= \int_0^{\pi/2} \frac{1}{\sqrt{1+\tan\left(\frac{\pi}{2}-x\right)}} dx \\ &= \int_0^{\pi/2} \frac{1}{\sqrt{1+\cot x}} dx \\ &= \int_0^{\pi/2} \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}} dx \quad \dots (ii) \end{aligned}$$

$\therefore$ From Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} 1 dx = [x]_0^{\pi/2} = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Question51

$$\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx =$$

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Options:

A. 0

B. $\frac{\pi}{2}$

C. $\frac{\pi^2}{2}$

D. $\frac{\pi^2}{4}$

Answer: D

Solution:

To solve the integral $I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$, we can use symmetry properties of definite integrals. According to the property:

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

Apply this property to transform the integral:

$$I = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

Adding both expressions for I , we have:

$$2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

Now, perform substitution to evaluate the new integral. Let $\cos x = t$, then $-\sin x dx = dt$. The limits of integration change as x changes from 0 to π , causing t to vary from 1 to -1. Thus, we have:

$$-\pi \int_1^{-1} \frac{dt}{1 + t^2}$$

Reversing the limits, the expression becomes:

$$\pi \int_{-1}^1 \frac{dt}{1 + t^2}$$

This integral evaluates to:

$$\pi [\tan^{-1} t]_{-1}^1$$

Simplifying:

$$= \pi (\tan^{-1}(1) - \tan^{-1}(-1))$$

Since $\tan^{-1}(1) = \frac{\pi}{4}$, it follows that:

$$= \pi \left(\frac{\pi}{4} + \frac{\pi}{4} \right) = \pi \times \frac{\pi}{2} = \frac{\pi^2}{2}$$

Thus, the original integral I is:

$$I = \frac{1}{2} \times \frac{\pi^2}{2} = \frac{\pi^2}{4}$$

Question52

$$\int_{-\pi}^{\pi} \frac{x \sin^3 x}{4 - \cos^2 x} dx =$$

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Options:

A. $2\pi(1 - \log 3)$

B. $2\pi \left(1 - \frac{3}{4} \log 3\right)$

C. $\pi \left(1 - \frac{3}{4} \log 3\right)$

D. $4\pi(1 - \log 3)$

Answer: B

Solution:

We have $I = \int_{-\pi}^{\pi} \frac{x \sin^3 x}{4 - \cos^2 x} dx$

Let $f(x) = \frac{x \sin^3 x}{4 - \cos^2 x}$

$$f(-x) = \frac{-x \times \sin^3(-x)}{4 - \cos^2(-x)} = \frac{x \sin^3 x}{4 - \cos^2 x} = f(x)$$

$\therefore f(x)$ is an even function

$$I = 2 \int_0^{\pi} \frac{x \sin^3 x}{4 - \cos^2 x} dx$$

$$I = 2 \int_0^{\pi} \frac{(\pi - x) \sin^3 x}{4 - \cos^2 x} dx$$

On adding Eqs. (i) and (ii), we get

$$2I = 2\pi \int_0^{\pi} \frac{\sin^3 x}{4 - \cos^2 x} dx$$

$$I = \pi \int_0^{\pi} \frac{\sin^3 x}{4 - \cos^2 x} dx$$

Let, $\cos x = t \Rightarrow -\sin x dx = dt$

When $x \rightarrow 0, t \rightarrow 1$ and $x \rightarrow \pi, t \rightarrow -1$

$$\begin{aligned}
 I &= -\pi \int_1^{-1} \frac{(1-t^2)}{4-t^2} dt \\
 &= \pi \int_{-1}^1 \left(\frac{-3}{4-t^2} + 1 \right) dt \\
 &= \pi \left[\frac{-3}{4} \log \left| \frac{2+t}{2-t} \right| + t \right]_{-1}^1 \\
 &= \pi \left[\left(\frac{-3}{4} \log 3 + 1 \right) - \left(-\frac{3}{4} \log \frac{1}{3} - 1 \right) \right] \\
 &= \pi \left[-\frac{3}{4} \log 3 + 1 - \frac{3}{4} \log 3 + 1 \right] \\
 &= 2\pi \left[1 - \frac{3}{4} \log 3 \right]
 \end{aligned}$$

Question53

$$\int_{-3}^3 |2-x| dx =$$

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Options:

- A. 12
- B. 16
- C. 13
- D. 25

Answer: C

Solution:

We have $I = \int_{-3}^3 |2 - x| dx$

$$|2 - x| = \begin{cases} (2 - x) & -3 < x < 2 \\ (x - 2) & 2 \leq x < 3 \end{cases}$$

$$\int_{-3}^3 |2 - x| dx = \int_{-3}^2 (2 - x) dx + \int_2^3 (x - 2) dx$$

$$= \left[2x - \frac{x^2}{2} \right]_{-3}^2 + \left[\frac{x^2}{2} - 2x \right]_2^3$$

$$= (4 - 2) - \left(-6 - \frac{9}{2} \right) + \left(\frac{9}{2} - 6 \right) - (2 - 4)$$

$$= 2 + 6 + \frac{9}{2} + \frac{9}{2} - 6 + 2 = 13$$

Question 54

$$\int_{\sqrt[5]{\frac{1}{31}}}^{\sqrt[5]{\frac{1}{242}}} \frac{1}{\sqrt[5]{x^{30} + x^{25}}} dx =$$

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Options:

A. $\frac{65}{4}$

B. $\frac{-75}{4}$

C. $\frac{75}{4}$

D. $\frac{-65}{4}$

Answer: D

Solution:

To solve the integral, we have:

$$I = \int_{\frac{1}{\sqrt[5]{31}}}^{\frac{1}{\sqrt[5]{242}}} \frac{1}{\sqrt[5]{x^{30} + x^{25}}} dx$$

First, we can rewrite the integrand as:

$$\int_{\frac{1}{\sqrt[5]{31}}}^{\frac{1}{\sqrt[5]{242}}} \frac{1}{x^6 \sqrt[5]{1 + \frac{1}{x^5}}} dx$$

To simplify, we perform a substitution. Let:

$$1 + \frac{1}{x^5} = t^5$$

Differentiating both sides gives:

$$-\frac{5}{x^6} dx = 5t^4 dt$$

Thus, the differential becomes:

$$dx = -x^6 \cdot \frac{1}{5t^4} dt$$

We need to adjust the limits of integration. When $x \rightarrow \frac{1}{\sqrt[5]{31}}$, then $t \rightarrow 2$. When $x \rightarrow \frac{1}{\sqrt[5]{242}}$, then $t \rightarrow 3$.

The integral now becomes:

$$I = - \int_2^3 \frac{1}{t} \times t^4 dt$$

This simplifies to:

$$I = - \int_2^3 t^3 dt$$

$$I = - \left[\frac{t^4}{4} \right]_2^3$$

Calculating the definite integral gives:

$$I = - \left(\frac{3^4}{4} - \frac{2^4}{4} \right)$$

$$I = - \left(\frac{81}{4} - \frac{16}{4} \right)$$

$$I = - \left(\frac{65}{4} \right)$$

Thus, the evaluated integral is:

$$I = \frac{-65}{4}$$

Question55

If $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{4}{n^2}\right) \left(1 + \frac{9}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{\frac{1}{n}} = ae^b$, then $a + b =$

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Options:

A. $\pi - 2$

B. π

C. $\pi + 2$

D. $\frac{\pi}{2}$

Answer: D

Solution:

Let,

$$P = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{4}{n^2}\right) \left(1 + \frac{9}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$$

$$\log P = \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \left(1 + \frac{1}{n^2}\right) + \log \left(1 + \frac{2^2}{n^2}\right) + \dots \log \left(1 + \frac{n^2}{n^2}\right) \right]$$

$$\log P = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2}\right)$$

$$= \int_0^1 \log(1+x^2) dx$$

$$= [x \log(1+x^2)]_0^1 - \int_0^1 \frac{2x^2}{1+x^2} dx$$

$$= \log 2 - 2 \int_0^1 dx + 2 \int_0^1 \frac{x^2 + 1 - 1}{1+x^2} dx$$

$$= \log 2 - 2 \int_0^1 dx + 2 \int_0^1 \frac{dx}{1+x^2}$$

$$= \log 2 - 2x \Big|_0^1 + 2 [\tan^{-1} x]_0^1$$

$$\log P = \log 2 - 2 + \frac{\pi}{2}$$

$$P = e^{\log 2 - 2 + \frac{\pi}{2}} = e^{\log 2} \cdot e^{-2 + \frac{\pi}{2}}$$

$$P = 2e^{-2 + \frac{\pi}{2}}$$

$$\therefore a = 2 \quad b = -2 + \frac{\pi}{2}$$

$$a + b = 2 - 2 + \frac{\pi}{2} = \frac{\pi}{2}$$

Question 56

$$\int_0^\pi x \sin^4 x \cos^6 x dx =$$

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Options:

A. $\frac{3\pi^2}{512}$

B. $\frac{3\pi^2}{256}$

C. $\frac{\pi^2}{256}$

D. $\frac{\pi^2}{512}$

Answer: A

Solution:

$$\text{Let } I = \int_0^\pi x \sin^4 x \cos^6 x dx \dots (i)$$

$$I = \int_0^\pi (\pi - x) \sin^4 x \cos^6 x dx \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$I = \pi \int_0^\pi \sin^4 x \times \cos^6 x dx$$

$$I = \frac{\pi}{2} \times 2 \int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x dx$$

$$I = \pi \int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x dx$$

$$= \frac{\pi \left[\frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \right]}{2 \times 120} = \frac{3\pi^2}{512}$$

Question 57

$$\text{If } I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx, \text{ then } I_{13} + I_{11} =$$

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Options:

A. $\frac{1}{13}$

B. $\frac{1}{12}$

C. $\frac{1}{10}$

D. $\frac{1}{11}$

Answer: B

Solution:

To determine I_n , we start with the given integral:

$$I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$$

Rewrite this as:

$$I_n = \int_0^{\frac{\pi}{4}} \tan^{n-2} x \times (\sec^2 x - 1) dx$$

Breaking it down, we have:

$$I_n = \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x dx - \int_0^{\frac{\pi}{4}} \tan^{n-2} x dx$$

Therefore, we can express this as:

$$I_n + I_{n-2} = \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x dx$$

When integrating, we find:

$$\int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x dx = \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\frac{\pi}{4}} = \frac{1}{n-1}$$

Hence, we deduce:

$$I_n + I_{n-2} = \frac{1}{n-1}$$

By setting $n = 13$, it follows that:

$$I_{13} + I_{11} = \frac{1}{12}$$

Thus, the sum $I_{13} + I_{11}$ is:

$$\frac{1}{12}$$

Question58

$$\lim_{n \rightarrow +\infty} \left[\frac{1}{n^4} + \frac{1}{(n^2+1)^{\frac{3}{2}}} + \frac{1}{(n^2+4)^{\frac{3}{2}}} + \frac{1}{(n^2+9)^{\frac{3}{2}}} + \dots + \frac{1}{4\sqrt{2}n^5} \right] =$$

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Options:

A. $\frac{3}{4\sqrt{2}}$

B. $\frac{3\sqrt{2}}{4}$

C. $\frac{5}{6\sqrt{2}}$

D. $\frac{5\sqrt{2}}{6}$

Answer: C

Solution:

$$\lim_{n \rightarrow \infty} x^4 \left[\frac{1}{x^5} + \frac{1}{(x^2+1)^{5/2}} + \frac{1}{(x^2+4)^{5/2}} \right. \\ \left. + \frac{1}{(x^2+9)^{5/2}} + \dots + \frac{1}{4\sqrt{2}x^5} \right]$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{x^4}{(x^2+r^2)^{5/2}} \left[\because \lim_{n \rightarrow \infty} \frac{x^4}{x^5} = 0 \right]$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \left[\frac{1}{\left[1 + \left(\frac{r}{n} \right)^2 \right]^{5/2}} \right]$$

$$= \int_0^1 \frac{1}{(1+x^2)^{5/2}} dx$$

$$\text{Let } x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$x \rightarrow 0, \theta \rightarrow 0 \text{ and}$$

$$x \rightarrow 1, \theta \rightarrow \frac{\pi}{4}$$

$$= \int_0^{\pi/4} \frac{1}{(1+\tan^2 \theta)^{5/2}} \times \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \frac{1}{\sec^3 \theta} d\theta$$

$$= \int_0^{\pi/4} \cos^3 \theta d\theta$$

$$= \frac{1}{4} \int_0^{\pi/4} (\cos 3\theta + 3 \cos \theta) d\theta$$

$$= \frac{1}{4} \left[\frac{\sin 3\theta}{3} + 3 \sin \theta \right]_0^{\pi/4}$$

$$= \frac{1}{4} \left[\frac{1}{3} \sin \left(\pi - \frac{\pi}{4} \right) + 3 \sin \frac{\pi}{4} \right]$$

$$= \frac{1}{4} \left[\frac{1}{3} \times \frac{1}{\sqrt{2}} + \frac{3}{\sqrt{2}} \right]$$

$$= \frac{1+9}{4 \times 3\sqrt{2}} \Rightarrow \frac{10}{4 \times 3\sqrt{2}} = \frac{5}{6\sqrt{2}}$$

Question 59

$$\int_{\log 4}^{\log 4} \frac{e^{2x} + e^x}{e^{2x} - 5e^x + 6} dx =$$

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Options:

A. $\log \left(\frac{64}{9} \right)$

B. $\log \left(\frac{256}{81} \right)$

C. $\log \left(\frac{32}{3} \right)$

D. $\log \left(\frac{128}{27} \right)$

Answer: D

Solution:

$$\begin{aligned} \text{Let } I &= \int_{\log 4}^{\log 5} \frac{e^{2x} + e^x}{e^{2x} - 5e^x + 6} dx \\ &= \int_{\log 4}^{\log 5} \frac{(e^x)^2 + e^x}{(e^x - 3)(e^x - 2)} dx \\ &= \int_{\log 4}^{\log 5} \frac{e^x (e^x + 1)}{(e^x - 3)(e^x - 2)} dx \end{aligned}$$

Let $e^x = t$

$\Rightarrow e^x dx = dt$

When $x \rightarrow \log 4, t \rightarrow 4$

$x \rightarrow \log 5, t \rightarrow 5$

$$\begin{aligned} I &= \int_4^5 \frac{t + 1}{(t - 3)(t - 2)} dt \\ &= \int_4^5 \frac{4}{t - 3} - \frac{3}{t - 2} \end{aligned}$$

[$\because$ by partial fraction]

$$\begin{aligned} &= [4 \log |t - 3| - 3 \log |t - 2|]_4^5 \\ &= \left[\log \left| \frac{(t - 3)^4}{(t - 2)^3} \right| \right]_4^5 \\ &= \log \frac{2^4}{3^3} - \log \frac{1}{2^3} = \log \frac{2^4}{3^3} \times 2^3 \\ &= \log \left(\frac{2^7}{3^3} \right) = \log \left(\frac{128}{27} \right) \end{aligned}$$

Question60

$$\int_1^2 \frac{x^4 - 1}{x^6 - 1} dx =$$

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Options:

A. $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$

B. $\frac{121}{6}$

C. $\sqrt{2} - 1$

D. $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$

Answer: A

Solution:

$$\begin{aligned} \text{Let } I &= \int_1^2 \frac{x^4 - 1}{x^6 - 1} dx \\ &= \int_1^2 \frac{(x^2 - 1)(x^2 + 1)}{(x^2 - 1)(x^4 + x^2 + 1)} dx \\ &= \int_1^2 \frac{x^2 + 1}{x^4 + x^2 + 1} dx = \int_1^2 \left(\frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2} + 1} \right) dx \\ &= \int_1^2 \left(\frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2} - 2 + 3} \right) dx \\ &= \int_1^2 \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x - \frac{1}{x}\right)^2 + 3} dx \end{aligned}$$

$$\text{Let } x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dt$$

When $x \rightarrow 1$, then $t \rightarrow 0$

When $x \rightarrow 2$, then $t \rightarrow \frac{3}{2}$

$$\begin{aligned} I &= \int_0^{3/2} \frac{1}{t^2 + 3} dt \\ &= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{1}{\sqrt{3}} \right]_0^{3/2} \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{3}{2\sqrt{3}} \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{\sqrt{3}}{2} \end{aligned}$$

Question61

Let $T > 0$ be a fixed number. $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that $f(x + T) = f(x), x \in \mathbb{R}$. If $I = \int_0^T f(x) dx$, then

$$\int_0^{5T} f(2x) dx =$$

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Options:

A. $10 I$

B. $\frac{5}{2} I$

C. $5 I$

D. $2 I$

Answer: C

Solution:

$$\text{Given, } I = \int_0^T f(x) dx$$

$$\text{If } f(x + T) = f(x)$$

$$\text{Now, } \int_0^{5T} f(2x) dx$$

On putting $2x = y$

$$\Rightarrow dx = \frac{1}{2} dy$$

$$\frac{1}{2} \int_0^{10T} f(y) dy = \frac{10I}{2} = 5I$$

Question62

$$\int_1^3 x^n \sqrt{x^2 - 1} dx = 6, \text{ then } n =$$

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Options:

A. 2

B. 3

C. 4

D. 5

Answer: B

Solution:

$$\begin{aligned} \text{Let } I &= \int_1^3 x \sqrt{x^2 - 1} dx = 6 \\ &= \frac{1}{2} \int_1^3 2x (x^2 - 1)^{\frac{1}{2}} dx = 6 \end{aligned}$$

$$\Rightarrow \left[\frac{(x^2 - 1)^{\frac{1}{2} + 1}}{\frac{1}{2} + 1} \right]_1^3 = 12$$

$$\Rightarrow \frac{(8)^{\frac{1}{2} + 1}}{\frac{1}{2} + 1} = 12$$

$$\Rightarrow \frac{(8)^{\frac{1}{2}}}{(n + 1)}(n) = \frac{3}{2}$$

Now, on putting $n = 3$, L.H.S = R. H. S

$\therefore n = 3$

Question63

$[\cdot]$ represents greatest integer function, then

$$\int_{-1}^1 (x[1 + \sin \pi x] + 1) dx =$$

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Options:

A. 1

B. 2

C. $\frac{5}{2}$

D. $\frac{3}{2}$

Answer: C

Solution:

$$\int_{-1}^1 (x[1 + \sin \pi x] + 1) dx$$

$$I = \int_{-1}^1 (x[1 + \sin \pi x] + 1) dx$$

We know that, $-1 \leq \sin \pi x \leq 1$

$$I = \int_{-1}^0 (x[1 + \sin \pi x] + 1) dx + \int_0^1 (x(1 + \sin \pi x) + 1) dx$$

$$\Rightarrow I = \int_{-1}^0 (x \cdot 0 + 1) dx + \int_0^1 (x + 1) dx$$

$$\Rightarrow I = [x]_{-1}^0 + \left[\frac{x^2}{2} + x \right]_0^1$$

$$\Rightarrow I = 0 - (-1) + \left[\frac{1}{2} + 1 \right]$$

$$\Rightarrow I = 2 + \frac{1}{2} = \frac{5}{2}$$

Question 64

$$\lim_{n \rightarrow \infty} \left[\frac{n}{(n+1)\sqrt{2n+1}} + \frac{n}{(n+2)\sqrt{2(2n+2)}} + \frac{n}{(n+3)\sqrt{3(2n+3)}} + \dots n \text{ terms} \right] = \int_0^1 f(x) dx$$

then $f(x) =$

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Options:

A. $\frac{1}{(1+x)\sqrt{x^2+2x}}$

B. $\frac{1}{(1+x)\sqrt{x+2}}$

C. $\frac{1}{(1+x)\sqrt{x^2+x+1}}$

D. $\frac{1}{(1+x)\sqrt{x^2-2x}}$

Answer: A

Solution:

$$\text{Here, } \lim_{n \rightarrow \infty} \left[\frac{n}{(n+1)\sqrt{2n+1}} + \frac{n}{(n+2)\sqrt{2(n+2)}} + \dots \text{ upto } n \text{ terms} \right]$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{n^2}{(n+k)\sqrt{k(2n+k)}} \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{n^2}{(n+k)\sqrt{k(2n+k)}} \times \frac{n^2}{n^2} \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\left(\frac{n+k}{n}\right)\sqrt{\frac{k}{n}\left(\frac{2n+k}{n}\right)}} \\
&= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\left(1 + \frac{k}{n}\right)\sqrt{\frac{k}{n}\left(2 + \frac{k}{n}\right)}} \\
&= \int_0^1 \frac{1}{(1+x)\sqrt{x(2-x)}} dx = \int_0^1 f(x) dx \\
f(x) &= \frac{1}{(1+x)\sqrt{2x+x^2}}
\end{aligned}$$

Question 65

If $I_n = \int_0^{\pi/4} \tan^n x dx$, then $\frac{1}{I_2+I_4} + \frac{1}{I_3+I_5} + \frac{1}{I_4+I_6} =$

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Options:

- A. $\frac{1}{I_9+I_{11}}$
- B. $\frac{1}{I_{10}+I_{12}}$
- C. $\frac{1}{I_{12}+I_{14}}$
- D. $\frac{1}{I_{11}+I_{13}}$

Answer: D

Solution:

$$\begin{aligned}
I_n &= \int_0^{\pi/4} \tan^n x dx \\
I_{n+2} &= \int_0^{\pi/4} \tan^{n+2} x dx
\end{aligned}$$

$$\begin{aligned}\text{Now, } I_n + I_{n+2} &= \int_0^{\frac{\pi}{4}} \tan^n x dx + \int_0^{\frac{\pi}{4}} \tan^n x \tan^2 x dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x (1 + \tan^2 x) dx = \int_0^{\frac{\pi}{4}} \tan^n x \sec^2 x dx\end{aligned}$$

$$\text{Put } \tan x = t \Rightarrow \sec^2 x dx = dt$$

$$\therefore I_n + I_{n+2} = \int_0^1 t^n dt = \frac{1}{n+1}$$

$$\begin{aligned}\text{Now, } \frac{1}{I_2 + I_4} + \frac{1}{I_3 + I_5} + \frac{1}{I_4 + I_6} \\ &= \frac{1}{1/3} + \frac{1}{1/4} + \frac{1}{1/5} \\ &= 3 + 4 + 5 = 12 = \frac{1}{I_{11} + I_{13}}\end{aligned}$$

Question66

$$\int_0^{\pi/4} e^{\tan^2 \theta} \sin^2 \theta \tan \theta d\theta =$$

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Options:

A. $\frac{1}{2} \left(\frac{e}{2} - 1 \right)$

B. $\frac{e}{2} - 1$

C. $\frac{\pi}{2}$

D. $2 \left(\frac{\pi}{2} - e \right)$

Answer: A

Solution:

$$\begin{aligned}\text{Let } I &= \int e^{\tan^2 \theta} \sin^2 \theta \tan \theta d\theta \\ &= \int e^{\tan^2 \theta} \times \sin^2 \theta \frac{\tan \theta \sec^2 \theta}{(1 + \tan^2 \theta)} d\theta \\ &= \frac{1}{2} \int 2e^{\tan^2 \theta} \times \frac{\tan^2 \theta}{(1 + \tan^2 \theta)} \times \frac{\tan \theta \sec^2 \theta}{(1 + \tan^2 \theta)} d\theta\end{aligned}$$

$$\text{Let } \tan^2 \theta = t$$

$$\Rightarrow 2 \tan \theta \sec^2 \theta d\theta = dt$$

When $\theta = 0$, then $t = 0$

and when $\theta = \frac{\pi}{4}$, then $t = 1$

$$\begin{aligned} \therefore I &= \frac{1}{2} \int_0^1 e^t \times \frac{t}{(1+t)^2} dt = \frac{1}{2} \int_0^1 e^t \left[\frac{1+t-1}{(1+t)^2} \right] dt \\ &= \frac{1}{2} \int_0^1 e^t \left[\frac{1}{1+t} + \left(\frac{-1}{(1+t)^2} \right) \right] dt \\ &= \frac{1}{2} \left[\frac{e^t}{1+t} \right]_0^1 \left[\because \int e^x (f(x) + f'(x) dx) = e^x f(x) + C \right] \\ &= \frac{1}{2} \left[\frac{e}{2} - 1 \right] \end{aligned}$$

Question67

$$\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) dt =$$

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Options:

A. 0

B. 1

C. 1/2

D. $\sqrt{3}/2$

Answer: A

Solution:

$$\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) dt$$

$$\begin{aligned}
&= \int_{\pi/4}^{5\pi/4} |\cos t| \sin t + \int_{\pi/4}^{5\pi/4} |\sin t| \cos t dt \\
&= \frac{2}{2} \int_{\pi/4}^{\pi/2} \cos \sin t dt + \frac{2}{2} \int_{\pi/2}^{5\pi/4} (-\cos t) \sin t dt \\
&\quad + \frac{2}{2} \int_{\pi/4}^{\pi} \sin t \cos t dt + \frac{2}{2} \int_{\pi}^{5\pi/4} (-\sin t) \cos t dt \\
&= \frac{1}{2} \int_{\pi/4}^{\pi/2} \sin 2t dt - \frac{1}{2} \int_{\pi/2}^{5\pi/4} \sin 2t dt + \frac{1}{2} \int_{\pi/4}^{\pi} \sin 2t dt \\
&\quad - \frac{1}{2} \int_{\pi}^{5\pi/4} \sin 2t dt \\
&= \frac{1}{2} \left[\frac{-\cos 2t}{2} \right]_{\pi/4}^{\pi/2} - \frac{1}{2} \left[\frac{-\cos 2t}{2} \right]_{\pi/2}^{5\pi/4} + \frac{1}{2} \left[\frac{-\cos 2t}{2} \right]_{\pi/4}^{\pi} \\
&\quad - \frac{1}{2} \left[\frac{-\cos 2t}{2} \right]_{\pi}^{5\pi/4} \\
&= \frac{-1}{4} \left(\cos \pi - \frac{\cos \pi}{2} \right) + \frac{1}{4} \left(\cos \frac{5\pi}{2} - \cos \pi \right) \\
&\quad - \frac{1}{4} \left(\cos 2\pi - \frac{\cos \pi}{2} \right) + \frac{1}{4} \left(\cos \frac{5\pi}{2} - \cos 2\pi \right) \\
&= \frac{-1}{4} (-1 - 0) + \frac{1}{4} (0 - (-1)) - \frac{1}{4} (1 - 0) + \frac{1}{4} (0 - 1) \\
&= \frac{1}{4} + \frac{1}{4} - \frac{1}{4} - \frac{1}{4} = 0
\end{aligned}$$

Question 68

If $f(x) = \max\{\sin x, \cos x\}$ and $g(x) = \min\{\sin x, \cos x\}$, then $\int_0^\pi f(x) dx + \int_0^\pi g(x) dx =$

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Options:

A. $2\sqrt{2} + 2$

B. $2\sqrt{2} - 2$

C. 2

D. $2\sqrt{2}$

Answer: C

Solution:

Given, $f(x) = \max\{\sin x, \cos x\}$ and $g(x) = \min\{\sin x, \cos x\}$

$$\begin{aligned}\therefore &= \int_0^\pi f(x)dx + \int_0^\pi g(x)dx = \int_0^\pi \sin x dx + \int_0^\pi \cos x dx \\ &= -[\cos x]_0^\pi + [\sin x]_0^\pi = -(-1 - 1) + 0 = 2\end{aligned}$$

Question 69

$$\int_0^1 a^k x^k dx =$$

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Options:

A. $\lim_{n \rightarrow \infty} \frac{a^k(1+2^k+3^k+\dots+n^k)}{n^{k+1}}$

B. $\lim_{n \rightarrow \infty} \frac{a^k+a^k+\dots+a^k}{n^{k+1}}$

C. $\lim_{n \rightarrow \infty} \frac{1}{n} \Sigma \left(\frac{r}{n} \right)^k$

D. $\lim_{n \rightarrow \infty} \frac{1}{n} \Sigma \left(\frac{2r}{n} \right)^k$

Answer: A

Solution:

If we consider the option (a), we have

$$\begin{aligned}\lim_{n \rightarrow \infty} & \frac{a^k (1 + 2^k + 3^k + \dots + n^k)}{n^k + 1} \\ &= \lim_{n \rightarrow \infty} \frac{a^k (1^k + 2^k + 3^k + \dots + n^k)}{n^k \cdot n} \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{a^k r^k}{n^k} \cdot \frac{1}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n a^k \left(\frac{r}{n} \right)^k \cdot \frac{1}{n}\end{aligned}$$

On putting $\frac{1}{n} \rightarrow dx, \frac{x}{n} \rightarrow x$ and $\Sigma \rightarrow \int$

$$= \int_0^1 a^k \cdot x^k \cdot dx$$

Question 70

Let α and β ($\alpha < \beta$) are roots of $18x^2 - 9\pi x + \pi^2 = 0$, $f(x) = x^2$, $g(x) = \cos x$. Then, $\int_{\alpha}^{\beta} x(g \circ f(x)) dx =$

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Options:

A. $\frac{\sqrt{3}-1}{4}$

B. $\frac{\sqrt{3}}{4}$

C. $\frac{2+\sqrt{3}}{2}$

D. $\frac{1}{2} \left(\sin \frac{\pi^2}{9} - \sin \frac{\pi^2}{36} \right)$

Answer: D

Solution:

$$f(x) = x^2, g(x) = \cos x$$

$$18x^2 - 9\pi x + \pi^2 = 0$$

$$\Rightarrow x = \frac{9\pi \pm \sqrt{81\pi^2 - 72\pi^2}}{36}$$

$$\Rightarrow x = \frac{9\pi \pm 3\pi}{36}$$

$$x = \frac{12\pi}{36}$$

$$\Rightarrow \beta = \frac{\pi}{3}$$

$$\text{and } x = \frac{6\pi}{36} \Rightarrow \alpha = \frac{\pi}{6} \quad [\because \alpha < \beta]$$

$$\text{Now, } g \circ f(x) = g(f(x)) = g(x^2) = \cos x^2$$

$$\begin{aligned}
\therefore \int_{\alpha}^{\beta} x(g \circ f(x)) dx &= \int_{\pi/6}^{\pi/3} x \cdot \cos x^2 dx \\
&= \frac{1}{2} \int_{\pi/6}^{\pi/3} \cos(x^2) \cdot 2x dx \\
&= \frac{1}{2} [\sin x^2]_{\pi/6}^{\pi/3} \\
&= \frac{1}{2} \left[\sin \frac{\pi^2}{9} - \sin \frac{\pi^2}{36} \right]
\end{aligned}$$

Question 71

$$\int_0^{\pi} x (\sin^2(\sin x) + \cos^2(\cos x)) dx =$$

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Options:

- A. π^2
- B. $\pi^2/2$
- C. 2π
- D. $\pi/4$

Answer: B

Solution:

$$\text{Let } I = \int_0^{\pi} x (\sin^2(\sin x) + \cos^2(\cos x)) dx \quad \dots(i)$$

$$I = \int_0^{\pi} (\pi - x) (\sin^2(\sin x) + \cos^2(\cos x)) dx \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \pi \int_0^{\pi} (\sin^2(\sin x) + \cos^2(\cos x)) dx$$

$$2I = 2\pi \int_0^{\pi/2} (\sin^2(\sin x) + \cos^2(\cos x)) dx$$

$$I = \pi \int_0^{\pi/2} \sin^2(\sin x) + \int_0^{\pi/2} \cos^2(\cos x) dx$$

$$I = \pi \int_0^{\pi/2} \sin^2(\cos x) + \int_0^{\pi/2} \cos^2(\cos x) dx$$

$$= \pi \int_0^{\pi/2} 1 dx$$

$$I = \pi \times \frac{\pi}{2}$$

$$I = \frac{\pi^2}{2}$$

Question 72

If $\int_0^{\pi} \log(\sin x) dx = 8k$, then $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$ is equal to

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Options:

A. k

B. $-k$

C. $\frac{k}{2}$

D. $4k$

Answer: B

Solution:

$$\int_0^{\pi} \log(\sin x) dx = 8k$$

$$\begin{aligned}
\text{Let } I &= \int_0^{\pi/4} \log(1 + \tan x) dx = \int_0^{\pi/4} \log(1 + \tan \theta) d\theta \\
&= \int_0^{\pi/4} \log(1 + \tan(\pi/4 - \theta)) d\theta \\
I &= \int_0^{\pi/4} \log\left(\frac{1 + \tan \theta + 1 - \tan \theta}{1 + \tan \theta}\right) d\theta \\
I &= \int_0^{\pi/4} \log 2 d\theta - \int_0^{\pi/4} \log(1 + \tan \theta) d\theta \\
I &= \int_0^{\pi/4} \log 2 d\theta - I \Rightarrow 2I = (\log 2) \left(\frac{\pi}{4}\right) \\
I &= \frac{(\log 2)\pi}{8} = \frac{-1}{4} \times \left(\frac{-\pi}{2} \log 2\right) = -\frac{1}{8}(-\pi \log 2) \\
&= \frac{-1}{8} \times 8k \quad \left[\begin{array}{l} \because \int_0^{\pi/2} \log \sin x dx = \frac{-\pi}{2} \log 2 \\ = \int_0^{\pi} \log \sin x = -\pi \log 2 \end{array} \right] \\
\therefore I &= -k
\end{aligned}$$

Question 73

If $\int_0^1 x^m (1-x)^n dx = k \int_0^1 x^n (1-x)^m dx$, then the value of k equals

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Options:

A. m

B. n

C. $\frac{1}{mn}$

D. 1

Answer: D

Solution:

$$\begin{aligned}\int_0^1 x^n(1-x)^n dx &= \int_0^1 (1-x)^n (1-(1-x)^n) dx \\ &= \int_0^1 (1-x)^n (x)^n dx \Rightarrow k = 1\end{aligned}$$

Question74

If $\int_0^a \frac{dx}{4+x^2} = \frac{\pi}{8}$, then the value of a is equal to

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Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

Solution:

$$\begin{aligned}\int_0^a \frac{dx}{4+x^2} &= \frac{\pi}{8} \\ \Rightarrow \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \Big|_0^a &= \frac{\pi}{8} \Rightarrow \tan^{-1} \left(\frac{a}{2} \right) = \frac{\pi}{4} \\ \Rightarrow a/2 = 1 &\Rightarrow a = 2\end{aligned}$$

Question75

$\int_1^2 \frac{x^3-1}{x^2}$ is equal to

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Options:

A. $\frac{5}{3}$

B. $\frac{3}{5}$

C. 1

D. -1

Answer: C

Solution:

$$\int_1^2 \left(\frac{x^3-1}{x^2} \right) dx = \frac{x^2}{2} + \frac{1}{x} \Big|_1^2 = \frac{5}{2} - \frac{3}{2} = 1$$

Question 76

If $\int_0^{\pi/2} \tan^n(x) dx = k \int_0^{\pi/2} \cot^n(x) dx$, then

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Options:

A. $k = 1$

B. $k = 2$

C. $k = \frac{1}{2}$

D. $k = 3$

Answer: A

Solution:

$$\begin{aligned}
\int_0^{\pi/2} \tan^n x dx &= k \int_0^{\pi/2} \cot^n(x) dx \\
\Rightarrow \int_0^{\pi/2} \tan^n \left(\frac{\pi}{2} - x \right) dx &= k \int_0^{\pi/2} \cot^n(x) dx \\
\left\{ \because \int_a^b f(x) dx &= \int_a^b f(a+b-x) dx \right\} \\
\Rightarrow \int_0^{\pi/2} \cot^n x dx &= k \int_0^{\pi/2} \cot^n x dx \\
k &= 1
\end{aligned}$$

Question77

$\int_0^2 x e^x dx$ is equal to

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Options:

- A. $e^2 + 1$
- B. $e^2 - 1$
- C. $e^{-1} - 1$
- D. $e^{-1} + 1$

Answer: A

Solution:

$$\begin{aligned}
\int_0^2 x e^x dx &= (x e^x - e^x)_0^2 \\
&= (2e^2 - e^2) - (0 \cdot e^0 - e^0) = e^2 + 1
\end{aligned}$$

Question78

$\int_2^4 \{|x - 2| + |x - 3|\} dx$ is equal to

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Options:

A. 1

B. 2

C. 3

D. 4

Answer: C

Solution:

$$\begin{aligned}\text{Let } I &= \int_2^4 (|x-2| + |x-3|) dx \\ \because x-a &= \begin{cases} x-a, & x \geq a \\ -(x-a), & x < a \end{cases} \\ \therefore I &= \int_2^4 |x-2| dx + \int_2^3 |x-3| dx + \int_3^4 |x-3| dx \\ &= \int_2^4 (x-2) dx - \int_2^3 (x-3) dx + \int_3^4 (x-3) dx \\ &= \left(\frac{x^2}{2} - 2x \right)_2^4 - \left(\frac{x^2}{2} - 3x \right)_2^3 + \left(\frac{x^2}{2} - 3x \right)_3^4 \\ &= \left[\left(\frac{16}{2} - 8 \right) - \left(\frac{4}{2} - 4 \right) \right] - \left[\left(\frac{9}{2} - 9 \right) - \left(\frac{4}{2} - 6 \right) \right] \\ &= \left| [0 - (-2)] - \left[\left(\frac{-9}{2} \right) - (-4) \right] + \left[(-4) - \left(-\frac{9}{2} \right) \right] \right| \\ &= \left| 2 + \frac{9}{2} - 4 - 4 + \frac{9}{2} \right| = [2 + 9 - 8] = 3\end{aligned}$$

Question 79

$$\int_{-1/2}^{1/2} \left\{ [x] + \log \left(\frac{1+x}{1-x} \right) \right\} dx \text{ is equal to}$$

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Options:

A. $2 \log(1/2)$

B. 0

C. $\frac{-1}{2}$

D. 1

Answer: C

Solution:

$$\begin{aligned} \text{(c) Let } I &= \int_{-1/2}^{1/2} \left\{ [x] + \log \left(\frac{1+x}{1-x} \right) dx \right\} \\ \therefore \int_{-a}^a f(x) dx &= \begin{cases} 2 \int_0^a f(x) dx, & f(-x) = f(x) \\ 0, & f(-x) = -f(x) \end{cases} \\ &= \int_{-1/2}^{1/2} [x] dx + \int_{-1/2}^{1/2} \log \left(\frac{1+x}{1-x} \right) dx \end{aligned}$$

$$\text{Let } f(x) = \log \left(\frac{1+x}{1-x} \right)$$

$$f(-x) = \log \left(\frac{1-x}{1+x} \right) = -f(x) \quad [\text{odd function}]$$

$$= \int_{-1/2}^{1/2} [x] dx + 0 = \int_{-1/2}^0 [x] dx + \int_0^{1/2} [x] dx$$

$$= \int_{-1/2}^0 (-1) dx + \int_0^{1/2} 0 dx$$

$$= -(x)_{-1/2}^0 + 0 = -\left(0 + \frac{1}{2}\right) = -\frac{1}{2}$$
